

FUNCTION

JEE EXERCISES

Exercise J-1

IMPORTANT FUNCTIONS (MODULUS, G.I.F., F.P.F. ETC.)

1. Let $f(x) = |x - 1|$, then :

- (A) $f(x^2) = (f(x))^2$ (B) $f(x+y) = f(x) + f(y)$ (C) $f(|x|) = |f(x)|$ (D) None of these

यदि $f(x) = |x - 1|$, तो :

- (A) $f(x^2) = (f(x))^2$ (B) $f(x+y) = f(x) + f(y)$ (C) $f(|x|) = |f(x)|$ (D) इनमें से कोई नहीं

Ans. D

Sol. $f(x) = |x - 1|$

$$\Rightarrow f(|x|) = ||x| - 1| = \begin{cases} |x| - 1 & \text{if } |x| \geq 1 \\ 1 - |x| & \text{if } |x| < 1 \end{cases}$$

$$\text{but } |f(x)| = ||x - 1|| = \begin{cases} x - 1 & \text{if } x \geq 1 \\ 1 - x & \text{if } x < 1 \end{cases}$$

$$\Rightarrow f(|x|) \neq |f(x)|$$

$$f(x+y) = |x+y-1| \neq |x-1| + |y-1|$$

2. If $f(x) = \cos \left[\frac{\pi^2}{2} \right] x + \sin \left[-\frac{\pi^2}{2} \right] x$, where $[x]$ denotes the greatest integer function, then which of the following is not correct :

यदि $f(x) = \cos \left[\frac{\pi^2}{2} \right] x + \sin \left[-\frac{\pi^2}{2} \right] x$, जहाँ $[x]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है, तो निम्नलिखित में से कौन-सा सही नहीं है :

- (A) $f(0) = 1$ (B) $f\left(\frac{\pi}{3}\right) = \frac{1}{\sqrt{3}+1}$ (C) $f\left(\frac{\pi}{2}\right) = 0$ (D) $f(\pi) = 0$

Ans. D

Sol. $f(x) = \cos \left[\frac{\pi^2}{2} \right] x + \sin \left[-\frac{\pi^2}{2} \right] x$

$$\Rightarrow f(x) = \cos 4x + \sin (-5x)$$

$$\Rightarrow f(0) = 1$$

$$f\left(\frac{\pi}{3}\right) = \cos \frac{4\pi}{3} - \sin \frac{5\pi}{3} = -\cos \frac{\pi}{3} + \sin \frac{\pi}{3}$$

$$= \frac{\sqrt{3}-1}{2} = \frac{1}{\sqrt{3}+1}$$

$$f\left(\frac{\pi}{2}\right) = \cos 2\pi - \sin\left(\frac{5\pi}{2}\right) = 0$$

$$f(\pi) = \cos 4\pi - \sin(5\pi) = 1$$

DOMAIN OF FUNCTIONS

3. Let $f(x)$ be a function whose domain is $[-5, 7]$. Let $g(x) = |2x + 5|$. Then domain of $(f \circ g)(x)$ is :

- (A) $[-4, 1]$ (B) $[-5, 1]$ (C) $[-6, 1]$ (D) None of these

माना $f(x)$ एक फलन है जिसका प्रान्त $[-5, 7]$ है। यदि $g(x) = |2x + 5|$ हो, तो $(f \circ g)(x)$ का प्रान्त है :

- (A) $[-4, 1]$ (B) $[-5, 1]$ (C) $[-6, 1]$ (D) इनमें से कोई नहीं

Ans. C

Sol. Domain of $f(g(x))$. Range of $g(x) \equiv$ Domain of $f(x)$

$$\Rightarrow -5 \leq |2x + 5| \leq 7 \Rightarrow 0 \leq |2x + 5| \leq 7 \Rightarrow -7 \leq 2x + 5 \leq 7$$

$$\Rightarrow -12 \leq 2x \leq 2 \Rightarrow -6 \leq x \leq 1$$

Hindi $f(g(x))$ के प्रान्त के लिए. $g(x)$ परिसर $\equiv f(x)$ प्रान्त

$$\Rightarrow -5 \leq |2x + 5| \leq 7 \Rightarrow 0 \leq |2x + 5| \leq 7 \Rightarrow -7 \leq 2x + 5 \leq 7$$

$$\Rightarrow -12 \leq 2x \leq 2 \Rightarrow -6 \leq x \leq 1$$

4. The domain of the function $f(x) = \log_{1/2} \left(-\log_2 \left(1 + \frac{1}{\sqrt[4]{x}} \right) - 1 \right)$ is :

- (A) $0 < x < 1$ (B) $0 < x \leq 1$ (C) $x \geq 1$ (D) null set

फलन $f(x) = \log_{1/2} \left(-\log_2 \left(1 + \frac{1}{\sqrt[4]{x}} \right) - 1 \right)$ का प्रान्त है :

- (A) $0 < x < 1$ (B) $0 < x \leq 1$ (C) $x \geq 1$ (D) रिक्त समुच्चय

Ans. D

Sol. $f(x) = \log_{1/2} \left(-\log_2 \left(1 + \frac{1}{\sqrt[4]{x}} \right) - 1 \right) \Rightarrow -\log_2 \left(1 + \frac{1}{x^{1/4}} \right) - 1 > 0$

$$\Rightarrow -\infty < \log_2 \left(1 + \frac{1}{x^{1/4}} \right) < -1 \Rightarrow 0 < 1 + \frac{1}{x^{1/4}} < \frac{1}{2} \Rightarrow -1 < \frac{1}{x^{1/4}} < -\frac{1}{2}$$

$$\Rightarrow x \in \phi \text{ (null set)} \Rightarrow x \in \phi \text{ (रिक्त समुच्चय)}$$

5. The domain of the function, $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ is :

फलन $f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$ का प्रान्त है :

- (A) $[1, 2]$ (B) $[2, 3]$ (C) $[2, 3]$ (D) $[1, 2]$

Ans. B

Sol. To define $f(x)$, $9 - x^2 > 0 \Rightarrow -3 < x < 3 \dots (1)$

$$-1 \leq (x-3) \leq 1 \Rightarrow 2 \leq x \leq 4 \dots (2)$$

From (i) and (ii), $2 \leq x < 3$ i.e., $[2, 3)$.

6. The domain of the function $f(x) = \frac{1}{\sqrt{|x|-x}}$ is : **[AIEEE-2011]**

फलन $f(x) = \frac{1}{\sqrt{|x|-x}}$ का प्रान्त है :

[AIEEE-2011]

- (A) $(-\infty, \infty)$ (B) $(0, \infty)$ (C) $(-\infty, 0)$ (D) $(-\infty, \infty) - \{0\}$

Ans. C

Sol. $f(x) = \frac{1}{\sqrt{|x|-x}}$

$$\Rightarrow |x| - x > 0$$

$$\Rightarrow |x| > x$$

$$\Rightarrow x \in (-\infty, 0)$$

RANGE OF FUNCTIONS

7. If $[2 \cos x] + [\sin x] = -3$, then the range of the function, $f(x) = \sin x + \sqrt{3} \cos x$ in $[0, 2\pi]$ is : (where $[\cdot]$ denotes greatest integer function)

यदि $[2 \cos x] + [\sin x] = -3$, हो तो फलन $f(x) = \sin x + \sqrt{3} \cos x$ का $[0, 2\pi]$ में परिसर है : (जहाँ $[\cdot]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है।)

- (A) $[-2, -1]$ (B) $(-2, -1]$ (C) $(-2, -1)$ (D) $[-2, -\sqrt{3})$

Ans. D

Sol. $[2 \cos x] + [\sin x] = -3$

$$-2 \leq 2 \cos x \leq 2 \Rightarrow [2 \cos x] = 2, 1, -1, -2$$

$$-1 \leq \sin x \leq 1 \Rightarrow [\sin x] = 1, 0, -1$$

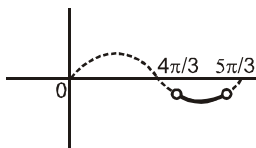
Equation holds true for $[2 \cos x] = -2$ and $[\sin x] = -1$

$$\Rightarrow -1 \leq \cos x < -\frac{1}{2} \quad \text{and} \quad -1 \leq \sin x < 0$$

$$\Rightarrow x \in \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right) \quad \text{and} \quad x \in (\pi, 2\pi) \Rightarrow x \in \left(\pi, \frac{4\pi}{3}\right)$$

$$f(x) = \sin x + \sqrt{3} \cos x$$

$$f(x) = 2 \sin \left(x + \frac{\pi}{3}\right)$$



$$\frac{4\pi}{3} < x + \frac{\pi}{3} < \frac{5\pi}{3} \Rightarrow -1 \leq \sin \left(x + \frac{\pi}{3}\right) < -\frac{\sqrt{3}}{2} \Rightarrow -2 \leq 2 \sin \left(x + \frac{\pi}{3}\right) < -\sqrt{3}$$

Hence range is $[-2, -\sqrt{3}]$

Hindi $[2 \cos x] + [\sin x] = -3$

$$-2 \leq 2 \cos x \leq 2 \Rightarrow [2 \cos x] = 2, 1, -1, -2$$

$$-1 \leq \sin x \leq 1 \Rightarrow [\sin x] = 1, 0, -1$$

समीकरण सत्य होगी यदि $[2 \cos x] = -2$ और $[\sin x] = -1$

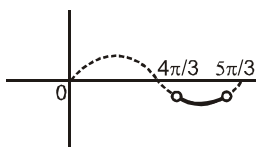
$$\Rightarrow -1 \leq \cos x < -\frac{1}{2} \quad \text{और} \quad -1 \leq \sin x < 0$$

$$\Rightarrow x \in \left(\frac{2\pi}{3}, \frac{4\pi}{3}\right) \quad \text{और} \quad x \in (\pi, 2\pi)$$

$$\Rightarrow x \in \left(\pi, \frac{4\pi}{3}\right)$$

$$f(x) = \sin x + \sqrt{3} \cos x$$

$$f(x) = 2 \sin \left(x + \frac{\pi}{3}\right)$$



$$\frac{4\pi}{3} < x + \frac{\pi}{3} < \frac{5\pi}{3} \Rightarrow -1 \leq \sin\left(x + \frac{\pi}{3}\right) < -\frac{\sqrt{3}}{2} \Rightarrow -2 \leq 2 \sin\left(x + \frac{\pi}{3}\right) < -\sqrt{3}$$

अतः परिसर है $y \in [-2, -\sqrt{3}]$.

8. Range of $f(x) = \log(3x^2 - 4x + 5)$ is :

फलन $f(x) = \log(3x^2 - 4x + 5)$ का परिसर है :

(A) $\left[\log \frac{11}{3}, \infty\right)$ (B) $[\log 10, \infty)$ (C) $\left[\log \frac{11}{6}, \infty\right)$ (D) $\left[\log \frac{11}{12}, \infty\right)$

Ans. A

Sol. $f(x) = \log_e(3x^2 - 4x + 5)$

$$3x^2 - 4x + 5 \geq \frac{11}{3} \Rightarrow \ln(3x^2 - 4x + 5) \geq \ln \frac{11}{3} \quad [\because \ln \text{ is an increasing function}]$$

$$\therefore \text{Range is } \left[\ln \frac{11}{3}, \infty\right)$$

Hindi $f(x) = \log_e(3x^2 - 4x + 5)$

$$3x^2 - 4x + 5 \geq \frac{11}{3} \Rightarrow \ln(3x^2 - 4x + 5) \geq \ln \frac{11}{3} \quad [\because \ln \text{ एक निरन्तर वर्द्धमान फलन है}]$$

$$\therefore \text{परिसर } \left[\ln \frac{11}{3}, \infty\right)$$

9. Range of $f(x) = 4^x + 2^x + 1$ is :

फलन $f(x) = 4^x + 2^x + 1$ का परिसर है :

(A) $(0, \infty)$ (B) $(1, \infty)$ (C) $(2, \infty)$ (D) $(3, \infty)$

Ans. B

Sol. $f(x) = 4^x + 2^x + 1$

Let $2^x = t > 0, \forall x \in \mathbb{R}$

$$\therefore f(x) = g(t) = t^2 + t + 1, t > 0$$

$$g(t) = \left(t + \frac{1}{2}\right)^2 + \frac{3}{4} \Rightarrow \left(t + \frac{1}{2}\right) > \frac{1}{2} \Rightarrow \left(t + \frac{1}{2}\right)^2 > \frac{1}{4} \Rightarrow \left(t + \frac{1}{2}\right)^2 + \frac{3}{4} > 1$$

Range is $(1, \infty)$

Hindi $f(x) = 4^x + 2^x + 1$

माना $2^x = t > 0, \forall x \in \mathbb{R}$

$$\therefore f(x) = g(t) = t^2 + t + 1, t > 0$$

$$g(t) = \left(t + \frac{1}{2}\right)^2 + \frac{3}{4} \Rightarrow \left(t + \frac{1}{2}\right) > \frac{1}{2} \Rightarrow \left(t + \frac{1}{2}\right)^2 > \frac{1}{4} \Rightarrow \left(t + \frac{1}{2}\right)^2 + \frac{3}{4} > 1$$

परिसर = $(1, \infty)$

10. The range of the functions $f(x) = \log_{\sqrt{2}}(2 - \log_2(16 \sin^2 x + 1))$ is:

फलन $f(x) = \log_{\sqrt{2}}(2 - \log_2(16 \sin^2 x + 1))$ का परिसर है :

- (A) $(-\infty, 1)$ (B) $(-\infty, 2)$ (C) $(-\infty, 1]$ (D) $(-\infty, 2]$

Ans. D

Sol. Here $(2 - \log_2(16 \sin^2 x + 1)) > 0 \Rightarrow 0 < 16 \sin^2 x + 1 < 4$

$$\Rightarrow \frac{-1}{16} \leq \sin^2 x < \frac{3}{16} \Rightarrow 1 \leq 16 \sin^2 x + 1 \leq 4 \Rightarrow 0 \leq \log_2(16 \sin^2 x + 1) < 2$$

$$\Rightarrow 2 \geq 2 - \log_2(16 \sin^2 x + 1) > 0$$

$$\Rightarrow \log_{\sqrt{2}} 2 \geq \log_{\sqrt{2}}(2 - \log_2(16 \sin^2 x + 1)) > -\infty \Rightarrow 2 \geq y > -\infty$$

Hence range is $y \in (-\infty, 2]$

Hindi यहाँ $(2 - \log_2(16 \sin^2 x + 1)) > 0 \Rightarrow 0 < 16 \sin^2 x + 1 < 4$

$$\Rightarrow \frac{-1}{16} \leq \sin^2 x < \frac{3}{16} \Rightarrow 1 \leq 16 \sin^2 x + 1 \leq 4$$

$$\Rightarrow 0 \leq \log_2(16 \sin^2 x + 1) < 2$$

$$\Rightarrow 2 \geq 2 - \log_2(16 \sin^2 x + 1) > 0$$

$$\Rightarrow \log_{\sqrt{2}} 2 \geq \log_{\sqrt{2}}(2 - \log_2(16 \sin^2 x + 1)) > -\infty$$

$$\Rightarrow 2 \geq y > -\infty$$

अतः परिसर है $y \in (-\infty, 2]$

11. The range of the function, $f(x) = \frac{2+x}{2-x}$, $x \neq 2$ is:

फलन $f(x) = \frac{2+x}{2-x}$, $x \neq 2$ का परिसर होगा :

- (A) $\mathbb{R}$ (B) $\mathbb{R} - \{-1\}$ (C) $\mathbb{R} - \{1\}$ (D) $\mathbb{R} - \{2\}$

Ans. B

Sol. $f(x) = \frac{x+2}{2-x}$; $x \neq 2$

$$\text{Range} = \mathbb{R} - \{-1\}$$

12. The range of the function, $f(x) = {}^{7-x}P_{x-3}$ is :

फलन $f(x) = {}^{7-x}P_{x-3}$ का परिसर है :

- (A) $\{1, 2, 3, 4\}$ (B) $\{1, 2, 3, 4, 5, 6\}$ (C) $\{1, 2, 3, 4, 5\}$ (D) $\{1, 2, 3\}$

Ans. D

Sol. $f(3) = {}^{7-3}P_0 = 1$, $f(4) = {}^3P_1 = 3$ and $f(5) = {}^2P_2 = 2$

Hence, range of $f = \{1, 2, 3\}$.

TRANSFORMATION OF GRAPHS

13. The graph of the function, $y = f(x)$ is symmetrical about the line $x = 2$, then :

यदि $y = f(x)$ का ग्राफ रेखा $x = 2$ के सापेक्ष सममित हो, तो :

- (A) $f(x) = f(-x)$ (B) $f(2+x) = f(2-x)$ (C) $f(x+2) = f(x-2)$ (D) $f(x) = -f(-x)$

Ans. B

Sol. $y = f(x)$ is symmetrical about the line $x = 2$

$$\Rightarrow f(2+x) = f(2-x)$$

CLASSIFICATION OF FUNCTIONS (ONE-ONE, ONTO ETC.)

14. If $f(x) = \frac{4a-7}{3}x^3 + (a-3)x^2 + x + 5$ is a one-one function, then :

- (A) $2 \leq a \leq 8$ (B) $1 \leq a \leq 2$ (C) $0 \leq a \leq 1$ (D) None of these

यदि $f(x) = \frac{4a-7}{3}x^3 + (a-3)x^2 + x + 5$ एकैकी फलन हो, तो :

- (A) $2 \leq a \leq 8$ (B) $1 \leq a \leq 2$ (C) $0 \leq a \leq 1$ (D) इनमें से कोई नहीं

Ans. A

Sol. $f(x) = \frac{4a-7}{3}x^3 + (a-3)x^2 + x + 5$

$$\frac{df}{dx} = \frac{4a-7}{3}x^2 + 2(a-3)x + 1$$

$$4a-7 > 0$$

$$4a > 7$$

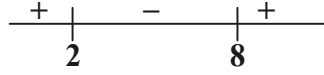
$$D = 4(a-3)^2 - 4(4a-7)$$

$$= 4(a^2 - 6a + 9 - 4a + 7)$$

$$\Rightarrow a > \frac{7}{4}$$

$$= 4(a^2 - 10a + 16)$$

$$= 4(a - 8)(a - 2)$$



$$\Rightarrow a \in [2, 8]$$

15. A function 'f' from the set of natural numbers to integers defined by, $f(x) = \begin{cases} \frac{n-1}{2} & \text{when 'n' is odd} \\ -\frac{n}{2} & \text{when 'n' is even} \end{cases}$ is :

- (A) onto but not one-one (B) one-one and onto both
(C) one-one but not onto (D) neither one-one nor onto

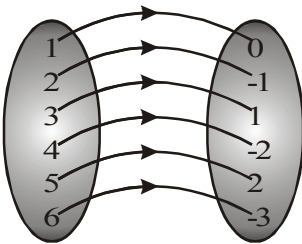
एक फलन 'f' प्राकृत संख्याओं के समुच्चय से पूर्णांकों के समुच्चय, $f(x) = \begin{cases} \frac{n-1}{2} & \text{, जहां 'n' विषम है} \\ -\frac{n}{2} & \text{, जहां 'n' सम है} \end{cases}$ में इस प्रकार है कि :

- (A) आच्छादक लेकिन एकैकी नहीं है (B) एकैकी एवं आच्छादक दोनों है
(C) एकैकी लेकिन आच्छादक नहीं है (D) न एकैकी न आच्छादक है

Ans. B

Sol. $f : \mathbb{N} \rightarrow \mathbb{I}$

$f(1) = 0, f(2) = -1, f(3) = 1, f(4) = -2, f(5) = 2$ and $f(6) = -3$ so on.



In this type of function every element of set A has unique image in set B and there is no element left in set B. Hence f is one-one and onto function.

16. If $f: \mathbb{R} \rightarrow \mathbb{S}$, defined by $f(x) = \sin x - \sqrt{3} \cos x + 1$, is onto, then the interval of 'S' is :

यदि $f: \mathbb{R} \rightarrow \mathbb{S}$ में परिभाषित फलन $f(x) = \sin x - \sqrt{3} \cos x + 1$ आच्छादक हो, तो 'S' का अन्तराल है :

- (A) $[0, 1]$ (B) $[-1, 1]$ (C) $[0, 3]$ (D) $[-1, 3]$

Ans. D

Sol. Using $-\sqrt{a^2 + b^2} \leq (a \sin x + b \cos x) \leq \sqrt{a^2 + b^2}$

$$-\sqrt{1 + (-\sqrt{3})^2} \leq (\sin x - \sqrt{3} \cos x) \leq \sqrt{1 + (-\sqrt{3})^2}$$

$$-2 \leq (\sin x - \sqrt{3} \cos x) \leq 2$$

$$-2 + 1 \leq (\sin x - \sqrt{3} \cos x + 1) \leq 2 + 1$$

$$-1 \leq (\sin x - \sqrt{3} \cos x + 1) \leq 3 \text{ i.e., range} = [-1, 3]$$

$\therefore$ For f to be onto $S = [-1, 3]$.

17. For real x , let $f(x) = x^3 + 5x + 1$, then :

[AIEEE-2009]

(A) f is one – one but not onto $\mathbb{R}$

(B) f is onto $\mathbb{R}$ but not one – one

(C) f is one – one and onto $\mathbb{R}$

(D) f is neither one – one nor onto $\mathbb{R}$

वास्तविक x के लिए $f(x) = x^3 + 5x + 1$ तो :

[AIEEE-2009]

(A) f समुच्चय $\mathbb{R}$ में एकेकी लेकिन आच्छादक नहीं

(B) f समुच्चय $\mathbb{R}$ में आच्छादक लेकिन एकेकी नहीं

(C) f समुच्चय $\mathbb{R}$ में एकेकी तथा आच्छादक

(D) f समुच्चय $\mathbb{R}$ में ना तो एकेकी ना ही आच्छादक

Ans. C

Sol. For real x , $f(x) = x^3 + 5x + 1$

$$\lim_{x \rightarrow \infty} f(x) = +\infty$$

$$\text{and } \lim_{x \rightarrow -\infty} f(x) = -\infty$$

$\therefore$ Range is $\mathbb{R}$ – $f(x)$ is on to

$$\text{Now } f'(x) = 3x^2 + 5 > 0$$

$\therefore f(x)$ is one-one

$f(x)$ is one-one onto.

18. The largest interval lying in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which the function, $\left[f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x) \right]$ is defined as :

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ में उपस्थित अधिकतम अन्तराल जिसके लिए फलन } \left[f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x) \right] \text{ परिभाषित}$$

है :

(A) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(B) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(C) $\left[0, \frac{\pi}{2}\right)$

(D) $[0, \pi]$

Ans. C

Sol. $f(x) = 4^{-x} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log \cos x$

$$\cos x > 0 \text{ \& } \frac{x}{2} - 1 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\ln \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ \& } x \in [2 - \pi, 2 + \pi]$$

$$x \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow x \in \left[0, \frac{\pi}{2}\right)$$

FUNCTIONAL EQUATION

19. If $f(x) = \cos(\ln x)$ then, $f(x)f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$ has the value :

- (A) -1 (B) $\frac{1}{2}$ (C) -2 (D) None of these

यदि $f(x) = \cos(\ln x)$ हो, तो $f(x)f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$ का मान है :

- (A) -1 (B) $\frac{1}{2}$ (C) -2 (D) इनमें से कोई नहीं

Ans. D

Sol. $f(x) = \cos(\ln x)$

$$f(x)f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$$

$$= \cos(\ln x)\cos(\ln y) - \frac{1}{2}[\cos(\ln x - \ln y) + \cos(\ln x + \ln y)]$$

$$= 0$$

20. If $y = f(x)$ satisfies the condition $f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$ ($x \neq 0$) then $f(x) =$

यदि $y = f(x)$ प्रतिबन्ध $f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}$ ($x \neq 0$) को सन्तुष्ट करता है, तो $f(x) =$

(A) $-x^2 + 2$ (B) $-x^2 - 2$ (C) $x^2 - 2, x \in \mathbb{R} - \{0\}$ (D) $x^2 - 2, |x| \in [2, \infty)$

Ans. D

Sol. $f\left(x + \frac{1}{x}\right) = \left(x + \frac{1}{x}\right)^2 - 2$ ($x \neq 0$)

$$x + \frac{1}{x} \leftrightarrow x$$

$$f(y) = x^2 - 2; x \in (-\infty, -2] \cup [2, \infty)$$

21. If $f(1) = 1$ and $f(n+1) = 2f(n) + 1$ if $n \geq 1$, then $f(n)$ is equal to :

यदि $f(1) = 1$ एवं $f(n+1) = 2f(n) + 1$ यदि $n \geq 1$, तो $f(n) =$

(A) $2^n + 1$ (B) 2^n (C) $2^n - 1$ (D) $2^{n-1} - 1$

Ans. C

Sol. $f(1) = 1$

$$f(n+1) = 2f(n) + 1$$

$$\Rightarrow f(2) = 2f(1) + 1 = 3 = 2^2 - 1$$

$$f(3) = 2f(2) + 1 = 7 = 2^3 - 1$$

$\vdots$

$$\Rightarrow f(n) = 2^n - 1$$

22. If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x+y) = f(x) + f(y)$, for all $x, y \in \mathbb{R}$ and $f(1) = 7$, then $\sum_{r=1}^n f(r)$ is :

यदि $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x+y) = f(x) + f(y)$, $\forall x, y \in \mathbb{R}$ को संतुष्ट करता है एवं $f(1) = 7$ हो, तो $\sum_{r=1}^n f(r)$ है :

(A) $\frac{7(n+1)}{2}$ (B) $7n(n+1)$ (C) $\frac{7n(n+1)}{2}$ (D) $\frac{7n}{2}$

Ans. C

Sol. $f(x+y) = f(x) + f(y) \quad \forall x, y \in \mathbb{R}$

$$\Rightarrow f(x) = kx$$

$$\text{Given } f(1) = 7$$

$$\Rightarrow k = 7$$

$$\Rightarrow \sum_{r=1}^n f(r) = \sum_{r=1}^n 7r = \frac{7 \cdot n(n+1)}{2}$$

23. If $f(x) + 2f\left(\frac{1}{x}\right) = 3x$, $x \neq 0$ and $S = \{x \in \mathbb{R} : f(x) = f(-x)\}$; then S : **[JEE(Main)-2016]**

(A) contains exactly two elements

(B) contains more than two elements

(C) is an empty set

(D) contains exactly one element

यदि $f(x) + 2f\left(\frac{1}{x}\right) = 3x$, $x \neq 0$ है, तथा $S = \{x \in \mathbb{R} : f(x) = f(-x)\}$ है; तो S : **[JEE(Main)-2016]**

(A) में तथ्यतः दो अवयव हैं

(B) में दो से अधिक अवयव हैं

(C) एक रिक्त समुच्चय है

(D) में केवल एक अवयव है

Ans. A

Sol. $f(x) + 2f(1/x) = 3x \quad (1) \quad (x \neq 0)$

$$f(1/x) + 2f(x) = \frac{3}{x} \quad (2)$$

From (1) & (2)

$$f(x) = \frac{2-x^2}{x}$$

$$f(x) = f(-x)$$

$$x^2 = 2$$

$$x = \pm \sqrt{2}$$

S contains exactly two elements

24. Let $a, b, c \in \mathbb{R}$. If $f(x) = ax^2 + bx + c$ is such that $a + b + c = 3$ and $f(x+y) = f(x) + f(y) + xy$, $\forall x, y \in \mathbb{R}$, then

$\sum_{n=1}^{10} f(n)$ is equal to : **[JEE(Main)-2017]**

माना $a, b, c \in \mathbb{R}$ | यदि $f(x) = ax^2 + bx + c$ ऐसा है कि $a + b + c = 3$ है तथा सभी $x, y \in \mathbb{R}$ के लिए

$f(x+y) = f(x) + f(y) + xy$ है, तो $\sum_{n=1}^{10} f(n)$ बराबर है :

[JEE(Main)-2017]

- (A) 190 (B) 255 (C) 330 (D) 165

Ans. C

Sol. $f(1) = 3$

Put $y = 1$ $f(x+1) - f(x) = 3 + x$

$(a(x+1)^2 + b(x+1) + c) - (ax^2 + bx + c) = 3 + x$

Comparing coefficients

$a = \frac{1}{2}$ $b = \frac{5}{2}$ $c = 0$

$f(n) = \frac{n^2 + 5n}{2}$

$\sum_{n=1}^{10} \frac{n^2 + 5n}{2} = 330$

Alternate Solution :

$f(1) = 3$

$f(2) = f(1) + f(1) + 1 \Rightarrow 3 + 3 + 1 = 7$

$f(3) = f(2) + f(1) + 2 \Rightarrow 7 + 3 + 2 = 12$

$f(4) = 18$

$f(5) = 25$

$f(6) = 33$

$f(7) = 42$

$f(8) = 52$

$f(9) = 63$

$f(10) = 75$

add = 330

COMPOSITE FUNCTIONS

25. If $f(x) = \frac{ax+b}{cx+d}$, then $(f \circ f)(x) = x$, provided that :

यदि $f(x) = \frac{ax+b}{cx+d}$ हो, तो $(f \circ f)(x) = x$, यह इस शर्त पर है कि :

- (A) $d + a = 0$ (B) $d - a = 0$ (C) $a = b = c = d = 1$ (D) $a = b = 1$

Ans. A

Sol. $f(x) = \frac{ax+b}{cx+d} \Rightarrow fof(x) = \frac{a\left(\frac{ax+b}{cx+d}\right)+b}{c\left(\frac{ax+b}{cx+d}\right)+d} \Rightarrow fof(x) = \frac{a^2x+ab+bcx+bd}{acx+bc+cdx+d^2}$

$$fof(x) = \frac{(a^2+bc)x+(ab+bd)}{(ac+cd)x+(bc+d^2)} = x$$

on comparing coefficient of both side $(a^2+bc)x+(ab+bd) = (ac+cd)x^2+(bc+d^2)x$

$$a^2+bc = bc+d^2 \Rightarrow a = d \text{ or } a = -d$$

$$\text{and } ab+bd = 0 \Rightarrow b = 0 \text{ or } a = -d$$

$$\text{and } ac+cd = 0 \Rightarrow c = 0 \text{ or } a = -d$$

which can be simultaneously true for $a = -d$

Hindi $f(x) = \frac{ax+b}{cx+d} \Rightarrow fof(x) = \frac{a\left(\frac{ax+b}{cx+d}\right)+b}{c\left(\frac{ax+b}{cx+d}\right)+d} \Rightarrow fof(x) = \frac{a^2x+ab+bcx+bd}{acx+bc+cdx+d^2}$

$$fof(x) = \frac{(a^2+bc)x+(ab+bd)}{(ac+cd)x+(bc+d^2)} = x \Rightarrow (a^2+bc)x+(ab+bd) = (ac+cd)x^2+(bc+d^2)x$$

समीकरण के दोनों पक्षों में गुणांकों की तुलना करने पर

$$a^2+bc = bc+d^2 \Rightarrow a = d \text{ या } a = -d$$

$$\text{तथा } ab+bd = 0 \Rightarrow b = 0 \text{ या } a = -d$$

$$\text{तथा } ac+cd = 0 \Rightarrow c = 0 \text{ या } a = -d$$

ये सभी एक साथ सत्य होगी यदि $a = -d$

26. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is: [JEE-2011]

(A) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$

(B) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$

(C) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

(D) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

माना $f(x) = x^2$ तथा $g(x) = \sin x$ सभी $x \in \mathbb{R}$ के लिए तो x के मान ज्ञात कीजिए जो $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$ को संतुष्ट करें, जहाँ $(f \circ g)(x) = f(g(x))$: [JEE-2011]

(A) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$

(B) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$

(C) $\frac{\pi}{2} + 2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

(D) $2n\pi, n \in \{\dots, -2, -1, 0, 1, 2, \dots\}$

Ans. A

Sol. Given $f(x) = x^2$; $g(x) = \sin x$

$$f \circ g \circ g \circ f(x) = \sin^2(\sin x^2)$$

$$\text{and } g \circ g \circ f(x) = \sin(\sin x^2)$$

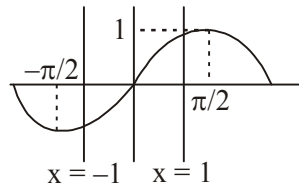
$$\text{given } f \circ g \circ g \circ f(x) = g \circ g \circ f(x)$$

$$\Rightarrow \sin^2(\sin x^2) = \sin(\sin x^2)$$

$$\Rightarrow \sin(\sin x^2) = 0 \text{ or } 1 \text{ (rejected)}$$

$$\sin(\sin x^2) = 0 \Rightarrow x^2 = n\pi$$

$$\Rightarrow x = \pm\sqrt{n\pi}; x \in \{0, 1, 2, 3, \dots\}$$



EVEN AND ODD FUNCTIONS

27. The function $f(x) = \log \left(\frac{1 + \sin x}{1 - \sin x} \right)$ is:

(A) Even

(B) Odd

(C) Neither even nor odd

(D) Both even and odd

$$\text{फलन } f(x) = \log \left(\frac{1 + \sin x}{1 - \sin x} \right) \text{ है:}$$

(A) सम फलन

(B) विषम फलन

(C) न तो सम न ही विषम

(D) सम एवं विषम दोनों

Ans. B

Sol. $f(x) = \log \left(\frac{1 + \sin x}{1 - \sin x} \right) \Rightarrow f(-x) = \log \left(\frac{1 - \sin x}{1 + \sin x} \right) = -\log \left(\frac{1 + \sin x}{1 - \sin x} \right) = -f(x)$ odd function विषम फलन

28. The function $f(x) = [x] + \frac{1}{2}$, $x \notin I$ is a/an:

(A) Even

(B) Odd

(C) Neither even nor odd

(D) None of these

$$\text{फलन } f(x) = [x] + \frac{1}{2}, x \notin I \text{ है:}$$

(A) सम

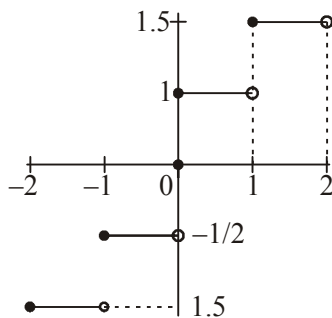
(B) विषम

(C) न सम न विषम

(D) इनमें से कोई नहीं

Ans. B

Sol. $f(x) = [x] + \frac{1}{2}, x \notin I \Rightarrow f(-x) = [-x] + \frac{1}{2} = -[x] - 1 + \frac{1}{2} = -\left([x] + \frac{1}{2}\right) = -f(x)$ odd function



INVERSE OF A FUNCTION

29. The inverse of the function $f(x) = y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ is :

फलन $f(x) = y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ का प्रतिलोम है :

- (A) $\frac{1}{2} \log \frac{1+x}{1-x}$ (B) $\frac{1}{2} \log \frac{2+x}{2-x}$ (C) $\frac{1}{2} \log \frac{1-x}{1+x}$ (D) $2 \log (1+x)$

Ans. A

Sol. $\frac{y}{1} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

By compendo and dividendo योगान्तरानुपात से

$$\frac{1+y}{1-y} = \frac{2e^x}{2e^{-x}} \Rightarrow 2x = \ln \left(\frac{1+y}{1-y} \right) \therefore f^{-1}(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

PERIODIC FUNCTIONS

30. Fundamental period of $f(x) = \sec(\sin x)$ is :

- (A) $\frac{\pi}{2}$ (B) 2π (C) π (D) aperiodic

$f(x) = \sec(\sin x)$ का मूलभूत आवर्तकाल है :

- (A) $\frac{\pi}{2}$ (B) 2π (C) π (D) अनावर्ती

Ans. C

Sol. $f(x) = \sec(\sin x)$. Since $\sin x$ is a periodic function with fundamental period 2π . $f(x)$ has a period 2π for fundamental period $f(x + \pi) = \sec(\sin(\pi + x)) = \sec(-\sin x) = \sec(\sin x) = f(x)$

$$f\left(x + \frac{\pi}{2}\right) \neq f(x) \text{ hence fundamental period is } \pi$$

Hindi. $f(x) = \sec(\sin x) \because \sin x$, मूलभूत आवर्तकाल 2π का आवर्ती फलन है, इसलिए $f(x)$ आवर्तकाल 2π है अब मूलभूत आवर्तकाल के लिए $f(x + \pi) = \sec(\sin(\pi + x)) = \sec(-\sin x) = \sec(\sin x) = f(x)$

$$f\left(x + \frac{\pi}{2}\right) \neq f(x) \text{ अतः मूलभूत आवर्तकाल } \pi \text{ है।}$$

31. The fundamental period of function $f(x) = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right] - 3x + 15$:

- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) 1 (D) Non-periodic

फलन $f(x) = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right] - 3x + 15$ का आवर्त काल होगा :

- (A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) 1 (D) आवर्ती फलन नहीं है

Ans. A

Sol. $f(x) = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right] - 3x + 15$

$$= \left\{ [x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + \dots + \left[x + \frac{n-1}{n}\right] = [nx] \right\} \quad n \in \mathbb{N}, x \in \mathbb{R}$$

$$\Rightarrow f(x) = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right] = [3x]$$

$$\Rightarrow f(x) = [3x] - 3x + 15$$

$$= -(3x - [3x]) + 15$$

$$= -\{3x\} + 15$$

$$\text{Periodic function with period} = \frac{1}{3}$$

Exercise J-2

IMPORTANT FUNCTIONS (MODULUS, SIGNUM, G.I.F., F.P.F. ETC.)

1. Read the symbols $[]$ and $\{ \}$ as greatest integer function less than or equal to x and fractional part function :

- (i) Find the number of real values of x , satisfying the equation $(x-2)[x] = \{x\} - 1$.
- (ii) Find the number of solutions of the equation, $x^2 - 3x + [x] = 0$ in the interval $[0, 3]$.
- (iii) If $[x]^2 + 3[x] - 10 \geq 0$, then find the range of x .
- (iv) If $[x]^2 - 5[x] + 6 = 0$, then find the range of x .

$[]$ व $\{ \}$ महत्तम पूर्णांक फलन व भिन्नात्मक फलन है :

- (i) समीकरण $(x-2)[x] = \{x\} - 1$ को सन्तुष्ट करने वाले x के वास्तविक मानों की संख्या ज्ञात कीजिए।
- (ii) अन्तराल $[0, 3]$ में समीकरण $x^2 - 3x + [x] = 0$ के हलों की संख्या होगी।
- (iii) यदि $[x]^2 + 3[x] - 10 \geq 0$ हो, तो x का परिसर ज्ञात कीजिए।
- (iv) यदि $[x]^2 - 5[x] + 6 = 0$ हो, तो x का परिसर ज्ञात कीजिए।

Ans. (i) infinite (ii) 2 (iii) $x \in (-\infty, -4) \cup [2, \infty)$ (iv) $x \in [2, 4)$

Sol. (i) $(x-2)[x] = \{x\} - 1$

$$\because \{x\} = x - [x]$$

$$\Rightarrow (x-2)[x] = x - [x] - 1$$

$$\Rightarrow x[x] - 2[x] = x - [x] - 1$$

$$\Rightarrow [x](x-1) = x-1$$

$$\Rightarrow (x-1)([x]-1) = 0$$

$$x=1 \text{ or } [x]=1$$

$$x=1 \text{ or } 0 \leq x < 1$$

$$\Rightarrow 0 \leq x \leq 1$$

Number of x satisfying given equation is infinite.

$$(ii) x^2 - 3x + [x] = 0; \quad x \in [0, 3]$$

$$\text{If } 0 \leq x < 1 \text{ then } x^2 - 3x = 0 \Rightarrow x = 0 \text{ or } x = 3$$

$$\Rightarrow x = 0$$

$$\text{If } 1 \leq x < 2 \text{ then } x^2 - 3x + 1 = 0 \Rightarrow x = \frac{3 \pm \sqrt{5}}{2}$$

$$\Rightarrow x \in \phi$$

$$\text{If } 2 \leq x < 3 \text{ then } x^2 - 3x + 2 = 0 \Rightarrow x = 2 \text{ or } 1$$

$$\Rightarrow x = 2$$

$\Rightarrow$ No. of values of x is 2.

$$(iii) [x]^2 + 3[x] - 10 \geq 0$$

$$\text{let } [x] = t$$

$$t^2 + 3t - 10 \geq 0$$

$$\Rightarrow (t + 5)(t - 2) \geq 0$$

$$\Rightarrow \begin{array}{c} + \quad | \quad - \quad | \quad + \\ -5 \quad \quad 2 \end{array}$$

$$\Rightarrow t \in (-\infty, -5] \cup [2, \infty)$$

$$\Rightarrow [x] \leq -5 \quad \text{or} \quad [x] \geq 2$$

$$\Rightarrow x < -4 \quad \text{or} \quad x \geq 2$$

$$\Rightarrow x \in (-\infty, -4) \cup [2, \infty)$$

$$(iv) [x]^2 - 5[x] + 6 = 0$$

$$\text{let } [x] = t$$

$$\Rightarrow t^2 - 5t + 6 = 0$$

$$\Rightarrow (t - 2)(t - 3) = 0$$

$$\Rightarrow [x] = 2 \quad \text{or} \quad [x] = 3$$

$$\Rightarrow 2 \leq x < 3 \quad \text{or} \quad 3 \leq x < 4$$

$$\Rightarrow x \in [2, 4)$$

DOMAIN AND RANGE OF FUNCTIONS

2. Find the domains of definitions of the following functions : (Read the symbols $[]$ and $\{\}$ as greatest integers and fractional part functions respectively.)

निम्न फलनों का प्रान्त ज्ञात कीजिये $\{[]\}$ व $\{\}$ महत्तम पूर्णांक फलन व भिन्नात्मक फलन है।)

$$(i) f(x) = \sqrt{\cos 2x} + \sqrt{16 - x^2}$$

$$(ii) f(x) = \log_7 \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x)$$

$$(iii) f(x) = \ln \left(\sqrt{x^2 - 5x - 24} - x - 2 \right)$$

$$(iv) f(x) = \sqrt{\frac{1 - 5^x}{7^{-x} - 7}}$$

$$(v) y = \log_{10} \sin(x - 3) + \sqrt{16 - x^2}$$

$$(vi) f(x) = \log_{100x} \left(\frac{2 \log_{10} x + 1}{-x} \right)$$

$$(vii) f(x) = \frac{1}{\sqrt{4x^2 - 1}} + \ln x(x^2 - 1)$$

$$(viii) f(x) = \sqrt{\log_{\frac{1}{2}} \frac{x}{x^2 - 1}}$$

$$(ix) f(x) = \sqrt{x^2 - |x|} + \frac{1}{\sqrt{9 - x^2}}$$

$$(x) f(x) = \sqrt{(x^2 - 3x - 10) \cdot \ln^2(x - 3)}$$

$$(xi) f(x) = \sqrt{\log_x(\cos 2\pi x)}$$

$$(xii) f(x) = \frac{\sqrt{\cos x - (1/2)}}{\sqrt{6 + 35x - 6x^2}}$$

$$(xiii) f(x) = \sqrt{\log_{1/3}(\log_4([x]^2 - 5))}$$

$$(xiv) f(x) = \frac{[x]}{2x - [x]} \quad (xv) f(x) = \log_x \sin x$$

$$(xvi) f(x) = \log_2 \left(-\log_{1/2} \left(1 + \frac{1}{\sin\left(\frac{x^\circ}{100}\right)} \right) \right) + \sqrt{\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3}$$

$$(xvii) f(x) = \frac{1}{[x]} + \log_{1 - \{x\}}(x^2 - 3x + 10) + \frac{1}{\sqrt{2 - |x|}} + \frac{1}{\sqrt{\sec(\sin x)}}$$

$$(xviii) f(x) = \sqrt{(5x - 6 - x^2) [\{ \ln \{x\} \}]} + \sqrt{(7x - 5 - 2x^2)} + \left(\ln \left(\frac{7}{2} - x \right) \right)^{-1}$$

$$(xix) f(x) = \log_{\left[\frac{x+1}{x} \right]} \left| x^2 - x - 6 \right| + {}^{16-x}C_{2x-1} + {}^{20-3x}P_{2x-5}$$

$$(xx) f(x) = \sqrt{\frac{\log_{0.3}(x-1)}{x^2 - 2x - 8}}, \quad g(x) = \sqrt{x^2 + 4x} C_{2x^2+3} \quad \text{and} \quad h(x) = \sqrt{1 - \log_3(x^2 - 6x + 11)}.$$

Ans. (i) $\left[-\frac{5\pi}{4}, \frac{-3\pi}{4} \right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4} \right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{4} \right]$ (ii) $\left(-4, -\frac{1}{2} \right) \cup (2, \infty)$ (iii) $(-\infty, -3]$

(iv) $(-\infty, -1) \cup [0, \infty)$ (v) $(3 - 2\pi < x < 3 - \pi) \cup (3 < x \leq 4)$

(vi) $\left(0, \frac{1}{100} \right) \cup \left(\frac{1}{100}, \frac{1}{\sqrt{10}} \right)$ (vii) $(-1 < x < -1/2) \cup (x > 1)$ (viii) $\left[\frac{1 - \sqrt{5}}{2}, 0 \right) \cup \left[\frac{1 + \sqrt{5}}{2}, \infty \right)$

(ix) $(-3, -1] \cup \{0\} \cup [1, 3)$ (x) $\{4\} \cup [5, \infty)$ (xi) $(0, 1/4) \cup (3/4, 1) \cup \{x : x \in \mathbb{N}, x \geq 2\}$

(xii) $\left(-\frac{1}{6}, \frac{\pi}{3} \right] \cup \left[\frac{5\pi}{3}, 6 \right)$ (xiii) $[-3, -2) \cup [3, 4)$ (xiv) $\mathbb{R} - \left\{ -\frac{1}{2}, 0 \right\}$

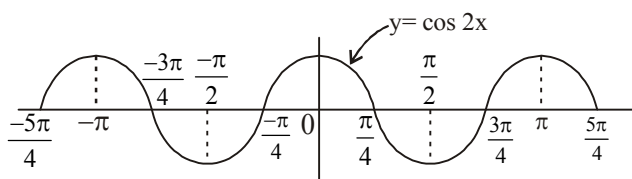
(xv) $2K\pi < x < (2K+1)\pi$ but $x \neq 1$ where K is non-negative integer

(xvi) $\{x \mid 1000 \leq x < 10000\}$ (xvii) $(-2, -1) \cup (-1, 0) \cup (1, 2)$ (xviii) $(1, 2) \cup (2, 5/2);$

(xix) $x \in \{4, 5\}$ (xx) $[2, 4); \{1, 2, 3\}; 2 \leq x \leq 4$

Sol. (i) $f(x) = \sqrt{\cos 2x} + \sqrt{16 - x^2}$

Domain $16 - x^2 \geq 0$ & $\cos x \geq 0$



$$\Rightarrow x \in \left[-\frac{5\pi}{4}, -\frac{3\pi}{4} \right] \cup \left[-\frac{\pi}{4}, \frac{\pi}{4} \right] \cup \left[\frac{3\pi}{4}, \frac{5\pi}{4} \right]$$

$$(ii) f(x) = \log_7 \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x)$$

$$\Rightarrow \log_5 \log_3 \log_2 (2x^3 + 5x^2 - 14x) > 0$$

$$\Rightarrow \log_3 \log_2 (2x^3 + 5x^2 - 14x) > 1$$

$$\Rightarrow 2x^3 + 5x^2 - 14x > 8$$

$$\Rightarrow 2x^3 + 5x^2 - 14x - 8 > 0$$

$$\Rightarrow (x-2)(2x^2 + 9x + 4) > 0$$

$$\Rightarrow (x-2)2\left(x + \frac{1}{2}\right)(x+4) > 0$$

$$\Rightarrow \begin{array}{c} - \quad + \quad - \quad + \\ | \quad | \quad | \quad | \\ -4 \quad -\frac{1}{2} \quad 2 \end{array}$$

$$\Rightarrow x \in \left(-4, -\frac{1}{2} \right) \cup (2, \infty)$$

$$(iii) f(x) = \ln \left(\sqrt{x^2 - 5x - 24} - x - 2 \right) \& \sqrt{x^2 - 5x - 24} > x + 2$$

$$\sqrt{x^2 - 5x - 24} > x + 2$$

$$x^2 - 5x - 24 > x^2 + 4x + 7$$

$$9x < -28$$

$$x < -\frac{28}{9} \quad \dots(1)$$

$$\text{Also, } x^2 - 5x - 24 \geq 0$$

$$(x-8)(x+3) \geq 0$$

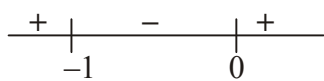
$$x \in (-\infty, -3] \cup [8, \infty) \quad \dots(2)$$

Intersection of (1) & (2)

$$x \in (-\infty, -3]$$

$$(iv) f(x) = \sqrt{\frac{1-5^x}{7^{-x}-7}}$$

$$\Rightarrow \frac{1-5^x}{7^{-x}-7} \geq 0 \quad \Rightarrow \frac{(5^x-1)7^x}{7^{x+1}-1} \geq 0$$

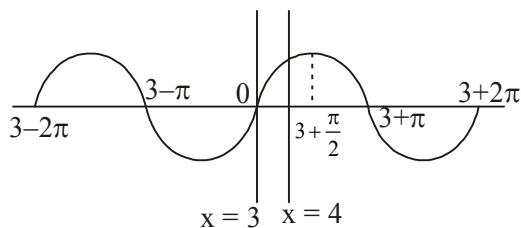
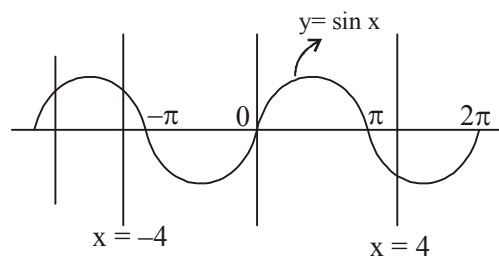


$$\Rightarrow x \in (-\infty, -1) \cup [0, \infty)$$

$$(v) y = \log_{10} \sin(x-3) + \sqrt{16-x^2}$$

$$\sin(x-3) > 0 \quad \& \quad 16-x^2 \geq 0 \quad \& \quad x \in [-4, 4]$$

$$\Rightarrow x \in (3-2\pi, 3-\pi) \cup (3, 4]$$

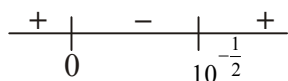


$$(vi) f(x) = \log_{100x} \left(\frac{2 \log_{10} x + 1}{-x} \right)$$

$$\Rightarrow 100x > 0 \quad \& \quad 100x \neq 1 \quad \& \quad \frac{2 \log_{10} x + 1}{-x} > 0$$

$$\Rightarrow x > 0 \quad \& \quad x \neq \frac{1}{100}$$

$$\Rightarrow x > 0 \quad \& \quad x \neq \frac{1}{100} \quad \& \quad \frac{2 \log_{10} x + 1}{x} < 0$$



$$\Rightarrow x > 0 \quad \& \quad x \neq \frac{1}{100} \quad \& \quad x \in \left(0, 10^{-\frac{1}{2}} \right)$$

$$\Rightarrow x \in \left(0, \frac{1}{100} \right) \cup \left(\frac{1}{100}, \frac{1}{\sqrt{10}} \right)$$

$$(vii) f(x) = \frac{1}{\sqrt{4x^2 - 1}} + \ln x (x^2 - 1)$$

$$x(x^2 - 1) > 0 \quad \& \quad 4x^2 - 1 > 0$$

$$\Rightarrow x(x-1)(x+1) > 0 \quad \& \quad x^2 - \frac{1}{4} > 0$$

$$\Rightarrow x \in (-1, 0) \cup (1, \infty) \quad \& \quad x \in \left(-\infty, -\frac{1}{2}\right) \cup \left(\frac{1}{2}, \infty\right)$$

$$\Rightarrow x \in \left(-1, -\frac{1}{2}\right) \cup (1, \infty)$$

$$(viii) f(x) = \sqrt{\log_{\frac{1}{2}} \frac{x}{x^2 - 1}}$$

$$\Rightarrow \log_{\frac{1}{2}} \frac{x}{x^2 - 1} \geq 0$$

$$\Rightarrow \frac{x}{x^2 - 1} \leq 1 \quad \&$$

$$\frac{x}{x^2 - 1} > 0$$

$$\begin{array}{c} - \quad + \quad - \quad + \\ | \quad | \quad | \quad | \\ -1 \quad 0 \quad 1 \end{array}$$

$$\frac{x - x^2 + 1}{x^2 - 1} \leq 0$$

$$\Rightarrow x \in (-1, 0) \cup (1, \infty)$$

$$\frac{x^2 - x - 1}{x^2 - 1} \geq 0$$

$$\begin{array}{c} + \quad - \quad + \quad - \quad + \\ | \quad | \quad | \quad | \quad | \\ -1 \quad \frac{1-\sqrt{5}}{2} \quad 1 \quad \frac{1+\sqrt{5}}{2} \end{array}$$

$$\Rightarrow x \in (-\infty, -1) \cup \left[\frac{1-\sqrt{5}}{2}, 1\right) \cup \left[\frac{1+\sqrt{5}}{2}, \infty\right)$$

$$\Rightarrow x \in \left[\frac{1-\sqrt{5}}{2}, 0\right) \cup \left[\frac{1+\sqrt{5}}{2}, \infty\right)$$

$$(ix) f(x) = \sqrt{x^2 - |x|} + \frac{1}{\sqrt{9 - x^2}}$$

$$x^2 - |x| \geq 0 \quad \&$$

$$9 - x^2 > 0$$

$$x^2 \leq |x| \quad \&$$

$$x^2 - 9 > 0$$

$$x \in (-\infty, -1] \cup [1, \infty) \cup \{0\} \quad \& \quad x \in (-3, 3)$$

$$x \in (-3, -1] \cup [1, 3) \cup \{0\}$$

$$(x) f(x) = \sqrt{(x^2 - 3x - 10) \ln^2(x - 3)}$$

$$(x^2 - 3x - 10) \ln^2(x - 3) \geq 0$$

$$\text{For } \ln^2(x - 3); x - 3 > 0 \text{ i.e. } x > 3$$

$$\Rightarrow x^2 - 3x - 10 \geq 0 \quad \text{or}$$

$$\ln^2(x - 3) = 0 \quad \rightarrow x \in \{4\}$$

$$\Rightarrow (x - 5)(x + 2) \geq 0$$

$$\Rightarrow x = 4$$

$$\Rightarrow x \in (-\infty, -2] \cup [5, \infty) \rightarrow x \in [5, \infty)$$

$$\rightarrow x \in [5, \infty) \cup \{4\}$$

$$(xi) f(x) = \sqrt{\log_x(\cos 2\pi x)}$$

$$\log_x(\cos 2\pi x) \geq 0 \quad \& \quad x > 0$$

$$\& \quad x \neq 1 \quad \& \quad \cos 2\pi x > 0$$

$$\Rightarrow \cos 2\pi x \geq 1 \quad \text{If } x > 1$$

$$\cos 2\pi x \leq 1 \text{ if } 0 < x < 1$$

$$\cos(2\pi x) = 1$$

$$x \in \mathbb{Z}$$

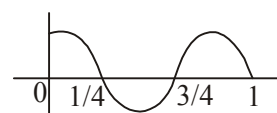
$$\text{Now } \cos(2\pi x) > 0$$

$$\text{If } 0 < x < 1 \text{ then } \cos 2\pi x > 0$$

$$x \in \left(0, \frac{1}{4}\right) \cup \left(\frac{3}{4}, 1\right)$$

final answer

$$x \in \left(0, \frac{1}{4}\right) \cup \left(\frac{3}{4}, 1\right) \cup \{x; x \in \mathbb{N}, x \geq 2\}$$



$$(xii) f(x) = \frac{\sqrt{\cos x - \frac{1}{2}}}{\sqrt{6 + 35x - 6x^2}}$$

Domain

$$\cos x \geq \frac{1}{2} \quad \&$$

$$-6x^2 + 35x + 6 < 0$$

$$\cos x \geq \frac{1}{2} \quad \&$$

$$6x^2 - 36x + x - 6 < 0$$

$$(x - 6)(6x + 1) < 0$$

$$\Rightarrow x \in \left(-\frac{1}{6}, 6\right)$$

$$(xiii) \ f(x) = \sqrt{\log_{\frac{1}{3}} \left(\log_4 \left([x]^2 - 5 \right) \right)}$$

$$\log_4 \left([x]^2 - 5 \right) > 0$$

$$[\mathbf{x}]^2 > 6$$

$$\Rightarrow [\mathbf{x}]^2 \leq 9$$

$$\Rightarrow 6 < [x]^2 \leq 9$$

$$\Rightarrow [\mathbf{x}]^2 = 9$$

$$\Rightarrow [x] = 3 \text{ or } [x] = -3$$

$$\Rightarrow x \in [-3, -2) \cup [3, 4)$$

$$(xiv) \quad f(x) = \frac{[x]}{2x - [x]}$$

Domain $2x - [x] \neq 0$

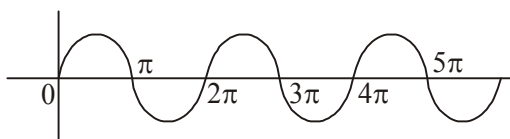
$$\Rightarrow 2x \neq [x]$$

$$2x = [x]$$

$$x \in \mathbb{R} - \left(\{0\} \cup \left\{ \frac{-1}{2} \right\} \right)$$

$$\Rightarrow x = 0 \text{ or } x = -\frac{1}{2}$$

$$\therefore \text{Domain} = \mathbb{R} - \left\{0, \frac{-1}{2}\right\}$$



(xv) $f(x) = \log_v \sin x$

$$x > 0 \text{ \& } x \neq 1 \text{ \& } \sin x > 0$$

$$\Rightarrow x \in (0, \pi) - \{1\} \cup (2\pi, 3\pi) \cup (4\pi, 5\pi) \cup \dots$$

$$x \in (2k\pi, (2k+1)\pi - \{1\}; k \in \mathbb{N} \cup \{0\})$$

$$(xvi) f(x) = \log_2 \left(-\log_{\frac{1}{2}} \left(1 + \frac{1}{\sin\left(\frac{x^\circ}{100}\right)} \right) \right) + \sqrt{\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3}$$

$$\text{For } \log_2 \left(-\log_{\frac{1}{2}} \left(1 + \frac{1}{\sin\left(\frac{x^\circ}{100}\right)} \right) \right)$$

$$\sin\left(\frac{x^\circ}{100}\right) \neq 0 \text{ \& } 1 + \frac{1}{\sin\left(\frac{x^\circ}{100}\right)} > 0 \text{ \& } -\log_{\frac{1}{2}} \left(1 + \frac{1}{\sin\left(\frac{x^\circ}{100}\right)} \right) > 0$$

$$\Rightarrow \sin\left(\frac{x^\circ}{100}\right) \neq 0 \text{ \& } \frac{1}{\sin\left(\frac{x^\circ}{100}\right)} > -1 \text{ \& } 1 + \frac{1}{\sin\left(\frac{x^\circ}{100}\right)} > 1$$

$$\Rightarrow \frac{1}{\sin\left(\frac{x^\circ}{100}\right)} > 0 \text{ \& } \sin\left(\frac{x^\circ}{100}\right) \neq 1$$

$$\Rightarrow \frac{x^\circ}{100} \in (2k\pi, (2k+1)\pi) \Rightarrow \frac{x}{1800} \in (2k, (2k+1))$$

$$\text{Now } \sqrt{\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3}$$

$$\log_{10}(\log_{10} x) - \log_{10}(4 - \log_{10} x) - \log_{10} 3 \geq 0$$

$$\text{\& } \log_{10} x > 0 \text{ \& } x > 0 \text{ \& } 4 - \log_{10} x > 0$$

$$\Rightarrow \log_{10} \left(\frac{\log_{10} x}{3 \cdot 4 - 3 \log_{10} x} \right) \geq 0$$

$$\Rightarrow \frac{4 \log_{10} x - 3 \cdot 4}{3 \cdot 4 - 3 \log_{10} x} \geq 0$$

$$\text{let } \log_{10} x = t$$

$$\frac{4t-12}{12-3t} \geq 0$$

$$\frac{4t-12}{3t-12} \leq 0$$

$$\Rightarrow t \in [3, 4)$$

$$\Rightarrow \log_{10} x \in [3, 4)$$

$$0 < \log_{10} x < 4$$

$$\Rightarrow \log_{10} x \in [3, 4)$$

$$\Rightarrow x \in [1000, 10000)$$

$$\Rightarrow x \in [1000, 10000)$$

$$(xvii) f(x) = \frac{1}{[x]} + \log_{1-\{x\}}(x^2 - 3x + 10)$$

$$+ \frac{1}{\sqrt{2-|x|}} + \frac{1}{\sqrt{\sec(\sin x)}}$$

$$[x] \neq 0 \quad 1 - \{x\} > 0 \text{ \& } x^2 - 3x + 10 > 0 \text{ \& } 1 - \{x\} \neq 1$$

$$\Rightarrow x \in \mathbb{R} - [0, 1) \quad \Rightarrow \{x\} < 1 \text{ \& } x \in \mathbb{R} \text{ \& } \{x\} \neq 0 \rightarrow x \notin \mathbb{Z}$$

$$2 - |x| > 0 \quad \sec(\sin x) > 0$$

$$\Rightarrow |x| < 2 \quad x \in \mathbb{R}$$

$$\Rightarrow -2 < x < 2$$

$$\Rightarrow x \in (-2, -1) \cup (-1, 0) \cup (1, 2)$$

$$(xviii) f(x) = \sqrt{(5x - 6 - x^2)[\{\ln\{x\}\}]} + \sqrt{(7x - 5 - 2x^2)} + \left(\ln\left(\frac{7}{2} - x\right)\right)^{-1}$$

For domain

$$7x - 5 - 2x^2 \geq 0 \quad \frac{7}{2} - x > 0 \text{ \& } \ln\left(\frac{7}{2} - x\right) \neq 0$$

$$2x^2 - 7x + 5 \leq 0 \quad \Rightarrow x < \frac{7}{2} \text{ \& } \frac{7}{2} - x \neq 1$$

$$x < \frac{7}{2} \text{ \& } \frac{5}{2} \neq x$$

$$\Rightarrow x \in \left[1, \frac{5}{2}\right] \quad x \in \left(-\infty, \frac{7}{2}\right) - \left\{\frac{5}{2}\right\}$$

$$x \in \left[1, \frac{5}{2}\right) \& \{x\} > 0$$

$$(5x - 6 - x^2) \underbrace{\left[\underbrace{\{\ln\{x\}\}}_{=0}\right]}_{=0} \geq 0 \& x \notin \mathbb{Z}$$

$$\Rightarrow x \in (1, 2) \cup \left(2, \frac{5}{2}\right) \text{ (final answer)}$$

$$(xix) f(x) = \log_{\left[x + \frac{1}{x}\right]} |x^2 - x - 6| + {}^{16-x}C_{2x-1} + {}^{20-3x}P_{2x-5}$$

$$\underbrace{x + \frac{1}{x}}_{\substack{\downarrow \\ x > 0}} > 0 \& \left[x + \frac{1}{x}\right] \neq 1 \& |x^2 - x - 6| \neq 0$$

$$x \neq -2 \quad x \neq 3 \\ \Rightarrow x \in (0, \infty) - \{3\} \quad \dots(1)$$

$$16 - x \in \mathbb{N} \& 2x - 1 \in \mathbb{N} \cup \{0\}$$

$$16 - x \geq 1 \& 2x - 1 \geq 0 \& 16 - x \geq 2x - 1$$

$$\Rightarrow \frac{1}{2} \leq x \leq \frac{17}{3} \quad \dots(2)$$

$$20 - 3x \in \mathbb{N} \& 2x - 5 \in \mathbb{N} \cup \{0\}$$

$$20 - 3x \geq 1 \& 2x - 5 \geq 0 \& 20 - 3x \geq 2x - 5$$

$$\frac{5}{2} \leq x \leq 5$$

$$x \in \{3, 4, 5\} \quad \dots(3)$$

$$\text{from (1), (2) \& (3)}$$

$$\Rightarrow x \in \{4, 5\}$$

$$(xx) f(x) = \sqrt{\frac{\log_{0.3}(x-1)}{x^2 - 2x - 8}}$$

$$\Rightarrow \frac{\log_{0.3}(x-1)}{x^2 - 2x - 8} \geq 0 \& x - 1 > 0$$

$$\Rightarrow x \in [2, 4)$$

$$g(x) = \sqrt{x^2 + 4x} C_{2x^2+3}$$

$$x^2 + 4x \in \mathbb{N} \& 2x^2 + 3 \in \mathbb{N} \cup \{0\}$$

$$\Rightarrow x^2 + 4x \geq 1 \quad \& \quad \underbrace{2x^2 + 3 \geq 0}_{x \in \mathbb{R}} \quad \& \quad \underbrace{x^2 + 4x \geq 2x^2 + 3}_{x^2 - 4x + 3 \leq 0}$$

$$\Rightarrow x^2 + 4x - 1 \geq 0$$

$$x \in (-\infty, -2 - \sqrt{5}] \cup [-2 + \sqrt{5}, \infty) \quad x \in [1, 3]$$

$$x \in \{1, 2, 3\}$$

$$h(x) = \sqrt{1 - \log_3(x^2 - 6x + 11)}$$

$$1 - \log_3(x^2 - 6x + 11) \geq 0 \quad \& \quad \underbrace{x^2 - 6x + 11 > 0}_{x \in \mathbb{R}}$$

$$\Rightarrow \log_3(x^2 - 6x + 11) \leq 1$$

$$\Rightarrow x^2 - 6x + 11 \leq 3$$

$$\Rightarrow x^2 - 6x + 8 \leq 0$$

$$\Rightarrow x \in [2, 4]$$

3. Find the domain & range of the following functions. (Read the symbols $[]$ and $\{\}$ as greatest integers and fractional part functions respectively.)

निम्न फलनों का प्रान्त व परिसर ज्ञात कीजिये। ($[]$ व $\{\}$ महत्तम पूर्णांक फलन व भिन्नात्मक फलन हैं।)

$$(i) \quad y = \log_{\sqrt{5}}(\sqrt{2}(\sin x - \cos x) + 3)$$

$$(ii) \quad y = \frac{2x}{1+x^2}$$

$$(iii) \quad f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6}$$

$$(iv) \quad f(x) = \frac{x}{1+|x|}$$

$$(v) \quad y = \sqrt{2-x} + \sqrt{1+x}$$

$$(vi) \quad f(x) = \log_{(\csc x - 1)}(2 - [\sin x] - [\sin x]^2) \quad (vii) \quad f(x) = \frac{\sqrt{x+4} - 3}{x-5}$$

- Ans.**
- (i) $D : x \in \mathbb{R} \quad R : [0, 2]$
 - (ii) $D = \mathbb{R}; \text{ range } [-1, 1]$
 - (iii) $D : \{x \mid x \in \mathbb{R}; x \neq -3; x \neq 2\} \quad R : \{f(x) \mid f(x) \in \mathbb{R}, f(x) \neq 1/5; f(x) \neq 1\}$
 - (iv) $D : \mathbb{R}; \quad R : (-1, 1)$
 - (v) $D : -1 \leq x \leq 2 \quad R : [\sqrt{3}, \sqrt{6}]$
 - (vi) $D : x \in (2n\pi, (2n+1)\pi) - \left\{2n\pi + \frac{\pi}{6}, 2n\pi + \frac{\pi}{2}, 2n\pi + \frac{5\pi}{6}, n \in \mathbb{I}\right\} \text{ and } R : \log_a 2; a \in (0, \infty) - \{1\} \Rightarrow \text{Range is } (-\infty, \infty) - \{0\}$
 - (vii) $D : [-4, \infty) - \{5\}; R : \left(0, \frac{1}{6}\right) \cup \left(\frac{1}{6}, \frac{1}{3}\right]$

Sol. (i) $y = \log_{\sqrt{5}} \left(\sqrt{2} (\sin x - \cos x) + 3 \right)$

for domain

$$\sqrt{2} (\sin x - \cos x) + 3 > 0$$

$$\Rightarrow \sin x - \cos x > \frac{-3}{\sqrt{2}}$$

$$\Rightarrow \sin x \cdot \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \cos x > -\frac{3}{2}$$

$$\Rightarrow \sin \left(x - \frac{\pi}{4} \right) > -\frac{3}{2}$$

$$x \in \mathbb{R}$$

Range

$$\sin x - \cos x \in [-\sqrt{2}, \sqrt{2}]$$

$$\Rightarrow \sqrt{2} (\sin x - \cos x) \in [-2, 2]$$

$$\Rightarrow 3 + \sqrt{2} (\sin x - \cos x) \in [1, 5]$$

$$\Rightarrow 1 \leq 3 + \sqrt{2} (\sin x - \cos x) \leq 5$$

$$\Rightarrow \log_{\sqrt{5}} 5 \geq y \geq 0$$

$$\Rightarrow 0 \leq y \leq 2$$

(ii) $y = \frac{2x}{1+x^2}$

Domain : $x \in \mathbb{R}$

For Range : $y + yx^2 - 2x = 0$

$$\Rightarrow x^2 \cdot y - 2x + y = 0$$

Case I : $y \neq 0$ D ≥ 0

$$4 - 4y^2 \geq 0$$

$$\Rightarrow y^2 - 1 \leq 0$$

$$\Rightarrow y \in [-1, 1] - \{0\}$$

Case II : $y = 0$

$$\Rightarrow x = 0 \text{ (Real)}$$

$$\Rightarrow y \in [-1, 1]$$

(iii) $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 6} = \frac{(x-1)(x-2)}{(x+3)(x-2)}$

Domain : $\mathbb{R} - \{2, -3\}$

$$\text{Range : } \mathbb{R} - \left\{1, \frac{1}{5}\right\}$$

$$(iv) f(x) = \frac{x}{1+|x|} = \begin{cases} \frac{x}{1-x} & x < 0 \\ \frac{x}{1+x} & x \geq 0 \end{cases}$$

$$\text{Domain} = \mathbb{R}$$

$$f(x) = \begin{cases} \frac{-x}{x-1} \equiv \frac{1-x-1}{x-1} = -1 - \frac{1}{x-1} & x < 0 \\ 1 - \frac{1}{1+x} & x \geq 0 \end{cases}$$

$$f(x) \in (-1, 1)$$

$$(v) y = \sqrt{2-x} + \sqrt{1+x}$$

$$\begin{aligned} \text{domain} \quad 2-x &\geq 0 & \& \quad 1+x \geq 0 \\ &\Rightarrow x \leq 2 & \& \quad x \geq -1 \end{aligned}$$

$$\Rightarrow x \in [-1, 2] \Rightarrow y^2 = 3 + 2\sqrt{2+2x-x-x^2}$$

$$= 3 + 2\sqrt{x-x^2+2}$$

$$\Rightarrow -x^2 + x + 2 \in \left[0, \frac{9}{4}\right]$$

$$\Rightarrow y^2 \in 3 + 2\left[0, \frac{3}{2}\right]$$

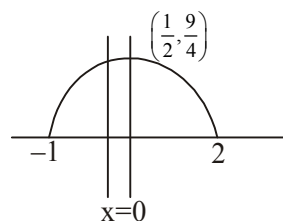
$$\Rightarrow y^2 \in [3, 6]$$

$$\Rightarrow y \in [\sqrt{3}, \sqrt{6}] \text{ as } y > 0$$

$$(vi) f(x) = \log_{\csc x - 1} \left(2 - [\sin x] - [\sin x]^2 \right)$$

$$\text{Domain}$$

$$\csc x - 1 > 0 \quad \& \quad \csc x \neq 2$$



$$\operatorname{cosec} x > 1 \quad \& \quad \operatorname{cosec} x \neq 2 \quad \rightarrow x \neq 2n\pi + \frac{\pi}{6}; n \in I$$

$$\& 2 - [\sin x] - [\sin x]^2 > 0$$

$$x \in (2n\pi, (2n+1)\pi) - \left\{ 2n\pi + \frac{\pi}{2}, 2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6} \right\}$$

$$\therefore \text{Domain} = (2n\pi, (2n+1)\pi) - \left\{ 2n\pi + \frac{\pi}{6}, 2n\pi + \frac{\pi}{2}, 2n\pi + \frac{5\pi}{6} \right\}; n \in I$$

Range

In this domain

$$\sin x \in (0, 1)$$

$$\Rightarrow [\sin x] = 0$$

$$\Rightarrow f(x) = \log_{\operatorname{cosec} x - 1} 2 \quad \Rightarrow \text{Range is } \mathbb{R} - \{0\}.$$

$$\operatorname{cosec} x - 1 > 0 \& \operatorname{cosec} x \neq 2$$

$$(vii) f(x) = \frac{\sqrt{x+4} - 3}{x-5}$$

$$\text{Domain} \quad x+4 \geq 0 \quad \& \quad x \neq 5$$

$$\Rightarrow x \in [-4, 5) \cup (5, \infty)$$

$$y = \frac{\sqrt{x+4} - 3}{x-5}$$

$$y = \frac{\sqrt{x+4} - 3}{(x+4) - 9}$$

$$\text{let } \sqrt{x+4} = t; t \geq 0$$

$$y = \frac{t-3}{t^2-9}$$

$$y = \frac{t-3}{(t-3)(t+3)}$$

$$y \in (0, \infty) - \left\{ -3, \frac{1}{6} \right\} \quad \text{and} \quad t+3 = \frac{1}{y}$$

$$t+3 \geq 3$$

$$\Rightarrow \frac{1}{y} \geq 3 \Rightarrow y \leq \frac{1}{3}$$

$$y \in \left(0, \frac{1}{6}\right) \cup \left[\frac{1}{6}, \frac{1}{3}\right]$$

4. Let $f(x) = (x+1)(x+2)(x+3)(x+4) + 5$ where $x \in [-6, 6]$. If the range of the function is $[a, b]$ where $a, b \in \mathbb{N}$ then find the value of $(a+b)$.

$f(x) = (x+1)(x+2)(x+3)(x+4) + 5$ जहाँ $x \in [-6, 6]$ यदि $f(x)$ का परिसर $[a, b]$ है $a, b \in \mathbb{N}$ तो $(a+b)$ का मान ज्ञात कीजिए।

Ans. 5049

Sol. $f(x) = (x+1)(x+2)(x+3)(x+4) + 5$
 $x \in [-6, 6]$

$$= (x^2 + 5x + 4)(x^2 + 5x + 6) + 5$$

Let $t = x^2 + 5x$

$$(t+4)(t+6) + 5 = t^2 + 10t + 24 + 5$$

$$= t^2 + 10t + 29$$

$$= (t+5)^2 + 4$$

$$f(x) = (x^2 + 5x + 5)^2 + 4$$

$$4 \leq f(x) \leq (36 + 30 + 5)^2 + 4$$

$$4 \leq f(x) \leq 5041 + 4 = 5045$$

$$a + b = 5049.$$

EQUAL OR IDENTICAL FUNCTIONS

5. Identify the pair(s) of functions which are identical? (where $[x]$ denotes greatest integer and $\{x\}$ denotes fractional part function)

(i) $f(x) = \operatorname{sgn}(x^2 - 3x + 4)$ and $g(x) = e^{[\{x\}]}$

(ii) $f(x) = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}}$ and $g(x) = \tan x$

(iii) $f(x) = \ln(1+x) + \ln(1-x)$ and $g(x) = \ln(1-x^2)$

(iv) $f(x) = \frac{\cos x}{1 - \sin x}$ and $g(x) = \frac{1 + \sin x}{\cos x}$

निम्न युग्मों में समरूप फलन ज्ञात करें? ($[]$ व $\{ \}$ महत्तम पूर्णांक फलन व भिन्नात्मक फलन है।)

(i) $f(x) = \operatorname{sgn}(x^2 - 3x + 4)$ तथा $g(x) = e^{[\{x\}]}$

$$(ii) f(x) = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} \text{ तथा } g(x) = \tan x$$

$$(iii) f(x) = \ln(1+x) + \ln(1-x) \text{ तथा } g(x) = \ln(1-x^2)$$

$$(iv) f(x) = \frac{\cos x}{1 - \sin x} \text{ तथा } g(x) = \frac{1 + \sin x}{\cos x}$$

Ans. (i), (iii) are identical

Sol. (i) $f(x) = \operatorname{sgn}(x^2 - 3x + 4) \equiv 1$ $a > 0$ & $D < 0$

$$g(x) = e^{\{x\}} \equiv 1 \Rightarrow f(x) = g(x)$$

$$D_f = \mathbb{R} \text{ \& } D_g = \mathbb{R}$$

$\Rightarrow f$ & g are identical

Range of $g(x)$ and $f(x) = \{1\}$

$$(ii) f(x) = \sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} \text{ and } g(x) = \tan x$$

$$D_g = \mathbb{R} - \left\{ (2n+1)\frac{\pi}{2}; n \in \mathbb{I} \right\}$$

$$D_f(x) = \mathbb{R} - \left\{ (2n+1)\frac{\pi}{2}; n \in \mathbb{I} \right\}$$

$$R_g = \mathbb{R} = (-\infty, \infty)$$

$$R_g = (0, \infty)$$

$$(iii) f(x) = \ln(1+x) + \ln(1-x)$$

$$g(x) = \ln(1-x^2)$$

$$D_f = (-1, 1)$$

$$f(x) = \ln(1-x^2); \quad g(x) = \ln(1-x^2)$$

$$D_g = (-1, 1)$$

$$\Rightarrow f(x) = g(x) \quad \forall x \in (-1, 1)$$

$\Rightarrow f$ & g are identical.

$$(iv) f(x) = \frac{\cos x}{1 - \sin x}$$

$$g(x) = \frac{1 + \sin x}{\cos x}$$

$$D_f = \mathbb{R} - \left\{ (4n+1)\frac{\pi}{2}; n \in \mathbb{I} \right\}$$

$$D_g = \mathbb{R} - \left\{ (2n+1)\frac{\pi}{2}; n \in \mathbb{I} \right\}$$

$$\Rightarrow D_f \neq D_g$$

$\Rightarrow f$ & g are not identical.

IMPLICIT AND EXPLICIT FUNCTIONS

6. (i) Write explicitly, functions of y defined by the following equations and also find the domains of definition of the given implicit functions :

y को x के पदों में लिखें। फलन का प्रान्त भी ज्ञात कीजिये। :

(a) $10^x + 10^y = 10$ (b) $x + |y| = 2y$

- (ii) The function $f(x)$ is defined on the interval $[0, 1]$. Find the domain of definition of the functions.

फलन $f(x)$ $[0, 1]$ में परिभाषित है। निम्न फलनों के प्रान्त ज्ञात कीजिये।

(a) $f(\sin x)$ (b) $f(2x+3)$

- (iii) Given that $y = f(x)$ is a function whose domain is $[4, 7]$ and range is $[-1, 9]$. Find the range and domain of :

दिया है $y = f(x)$ एक ऐसा फलन है जिसका प्रान्त $[4, 7]$ तथा परिसर $[-1, 9]$ है। निम्न फलनों के प्रान्त व परिसर ज्ञात कीजिये:

(a) $g(x) = \frac{1}{3} f(x)$ (b) $h(x) = f(x-7)$

Ans. (i) (a) $y = \log(10 - 10^x)$, $-\infty < x < 1$ (b) $y = x/3$ when $-\infty < x < 0$ and $y = x$ when $0 \leq x < +\infty$

(ii) (a) $2K\pi \leq x \leq 2K\pi + \pi$ where $K \in I$ (b) $[-3/2, -1]$

(iii) (a) Range : $[-1/3, 3]$, Domain = $[4, 7]$ (b) Range $[-1, 9]$ and domain $[11, 14]$

Sol. (i) (a) $10^x + 10^y = 10$

$$10^y = 10 - 10^x$$

$$y = \log_{10}(10 - 10^x)$$

$$10 - 10^x > 0$$

$$\Rightarrow 10^x < 10$$

$$\Rightarrow x < 1$$

$$\therefore \text{Domain} = (-\infty, 1)$$

(b) $x + |y| = 2y$

$$\begin{array}{l} \swarrow \quad \searrow \\ y \geq 0 \quad y < 0 \\ \swarrow \quad \searrow \\ x = y \quad x = 3y \end{array}$$

$$\therefore y = x \text{ if } 0 \leq x < \infty$$

$$y = x \text{ if } -\infty < x < 0$$

(ii) $f(x); x \in [0, 1]$

(a) $f(\sin x)$

$$\Rightarrow \sin x \in [0, 1]$$

$$\Rightarrow x \in [2k\pi, (2k+1)\pi]; k \in I$$

(b) $f(2x+3)$

$$\Rightarrow 2x + 3 \in [0, 1]$$

$$\Rightarrow 2x \in [-3, -2]$$

$$\Rightarrow x \in \left[-\frac{3}{2}, -1\right]$$

(iii) $f(x)$

$$D_f = [4, 7]$$

$$R_f = [-1, 9]$$

$$\Rightarrow g(x) = \frac{1}{3}f(x)$$

$$h(x) = f(x - 7)$$

$$\Rightarrow D_g = [4, 7]$$

$$D_h = [11, 14]$$

$$R_g = \left[-\frac{1}{3}, 3\right]$$

$$R_h = [-1, 9]$$

CLASSIFICATION OF FUNCTIONS (ONE-ONE, ONTO ETC.)

7. Classify the following functions $f(x)$ defined in $R \rightarrow R$ as injective, surjective, both or none :

निम्न फलनों में एकैकी, आच्छादक, दोनों या कोई भी नहीं ज्ञात करें। $f: R \rightarrow R$:

$$(a) f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$$

$$(b) f(x) = x^3 - 6x^2 + 11x - 6$$

$$(c) f(x) = (x^2 + x + 5)(x^2 + x - 3)$$

Ans. (a) neither surjective nor injective (b) surjective but not injective (c) neither injective nor surjective

Sol. (a) $f(x) = \frac{x^2 + 4x + 30}{x^2 - 8x + 18}$

Neither injective nor surjective.

$$(b) f(x) = x^3 - 6x^2 + \underbrace{11x - 6}_{\text{surjective}}$$

Neither injective as $f'(x) = 3x^2 - 12x + 11$

$$(c) f(x) = (x^2 + x + 5)(x^2 + x - 3)$$

Neither injective nor surjective

FUNCTIONAL EQUATION

8. Solve the following problems from (a) to (e) on functional equation.

फलनो के समीकरण पर आधारित प्रश्न (a) से (e) तक हल कीजिए।

- (a) The function $f(x)$ defined on the real numbers has the property that $f(f(x)) \cdot (1 + f(x)) = -f(x)$ for all x in the domain of f . If the number 3 is in the domain and range of f , compute the value of $f(3)$.

फलन $f(x)$ यह गुणधर्म है कि $f(f(x)) \cdot (1 + f(x)) = -f(x)$ जहाँ x प्रांत का अवयव है। यदि 3 प्रान्त व परिसर दोनों में है तो $f(3)$ ज्ञात कीजिए।

- (b) Suppose f is a real function satisfying $f(x + f(x)) = 4f(x)$ and $f(1) = 4$. Find the value of $f(21)$.

फलन f सम्बन्ध $f(x + f(x)) = 4f(x)$ को संतुष्ट करता है तथा $f(1) = 4$ । $f(21)$ का मान ज्ञात कीजिए।

- (c) Let ' f ' be a function defined from $\mathbb{R}^+ \rightarrow \mathbb{R}^+$. If $[f(xy)]^2 = x(f(y))^2$ for all positive numbers x and y and $f(2) = 6$, find the value of $f(50)$.

फलन ' f ' $\mathbb{R}^+ \rightarrow \mathbb{R}^+$ यदि $[f(xy)]^2 = x(f(y))^2$ x व y धनात्मक है तथा $f(2) = 6$ तो $f(50)$ का मान ज्ञात कीजिए।

- (d) Let $f(x)$ be a function with two properties

फलन $f(x)$ में दो निम्न गुणधर्म हैं

- (i) for any two real number x and y , $f(x + y) = x + f(y)$ and (ii) $f(0) = 2$.

Let $f(x)$ be a function with two properties

वास्तविक संख्या x व y के लिये $f(x + y) = x + f(y)$ तथा (ii) $f(0) = 2$.

Find the value of $f(100)$.

$f(100)$ का मान ज्ञात करें।

- (e) Let f be a function such that $f(3) = 1$ and $f(3x) = x + f(3x - 3)$ for all x . Then find the value of $f(300)$.

फलन f ऐसा है कि $f(3) = 1$ तथा $f(3x) = x + f(3x - 3)$, x के सभी वास्तविक मानों के लिये $f(300)$ का मान ज्ञात कीजिए।

- (f) Suppose that $f(x)$ is a function of the form $f(x) = \frac{ax^8 + bx^6 + cx^4 + dx^2 + 15x + 1}{x}$ ($x \neq 0$). If $f(5) = 2$ then find the value of $f(-5)$.

यदि फलन $f(x)$ ऐसा है कि $f(x) = \frac{ax^8 + bx^6 + cx^4 + dx^2 + 15x + 1}{x}$ ($x \neq 0$). व $f(5) = 2$ तो $f(-5)$ का मान ज्ञात कीजिए।

Ans. (a) $-3/4$ (b) 64 (c) 30 (d) 102 (e) 5050 (f) 28

Sol. (a) $f(f(x)) \cdot (1 + f(x)) = -f(x)$

$$\Rightarrow f(f(x)) = -\frac{f(x)}{1 + f(x)}$$

$$f(x) \leftrightarrow x$$

$$\Rightarrow f(x) = -\frac{x}{1 + x}$$

$$f(3) = -\frac{3}{4}$$

$$(b) f(x + f(x)) = 4 f(x) \text{ \& } f(1) = 4$$

$$\Rightarrow f(5) = 16$$

$$\therefore f(x) = 3x + 1$$

$$\Rightarrow f(21) = 63 + 1 = 64$$

$$(c) [f(xy)]^2 = x(f(y))^2$$

$$y = 2$$

$$\Rightarrow (f(2x))^2 = x \cdot 36$$

$$\Rightarrow (f(2x))^2 = 36x$$

$$\text{put } x = 25$$

$$\Rightarrow f(50) = 30$$

$$(d) f(x + y) = x + f(y)$$

$$\text{put } y = 0$$

$$f(x) = x + 2$$

$$\Rightarrow f(100) = 102$$

$$(e) f(3x) = x + f(3x - 3)$$

$$\Rightarrow f(3x) - f(3x - 3) = x \Rightarrow f(0) = 0$$

$$\Rightarrow \sum_{x=1}^{100} f(3x) - f(3x - 3) = \sum_{x=1}^{100} x = 5050$$

$$\Rightarrow -f(0) + f(300) = 5050$$

$$\Rightarrow f(300) = 5050$$

$$(f) f(x) = \frac{ax^8 + bx^6 + cx^4 + dx^2 + 15x + 1}{x}$$

$$f(5) + f(-5) = \frac{15(5+5)}{5}$$

$$2 + f(-5) = 30$$

$$\Rightarrow f(-5) = 28$$

COMPOSITE FUNCTIONS

9. Suppose $f(x) = \sin x$ and $g(x) = 1 - \sqrt{x}$. Then find the domain and range of the following functions :

$f(x) = \sin x$ व $g(x) = 1 - \sqrt{x}$ तो निम्न फलनों का प्रान्त व परिसर ज्ञात कीजिए :

- (a) fog (b) gof (c) fof (d) gog

- Ans.** (a) Domain is $x \geq 0$; range $[-1, 1]$ (b) Domain $2k\pi \leq x \leq 2k\pi + \pi$; range $[0, 1]$
(c) Domain $x \in \mathbb{R}$; range $[-\sin 1, \sin 1]$ (d) Domain is $0 \leq x \leq 1$; range is $[0, 1]$

Sol. $f(x) = \sin x$ $g(x) = 1 - \sqrt{x}$

(a) $\text{fog}(x) = \sin(1 - \sqrt{x})$

Domain = $[0, \infty)$

Range = $[-1, 1]$

(b) $\text{gof}(x) = g(\sin x) = 1 - \sqrt{\sin x}$

for domain

$\sin x \geq 0$

$x \in [2k\pi, (2k+1)\pi]; k \in \mathbb{I}$

$\therefore \sin x \in [-1, 1]$

$\Rightarrow -\sqrt{\sin x} \in [-1, 0]$

$\Rightarrow 1 - \sqrt{\sin x} \in [0, 1]$

(c) $\text{fof}(x) = \sin(\sin x)$

Domain = $(-\infty, \infty)$

Range = $[-\sin 1, \sin 1]$

(d) $\text{gog}(x) = g(1 - \sqrt{x})$

$= 1 - \sqrt{1 - \sqrt{x}}$

Domain : $x \geq 0$ & $1 - \sqrt{x} \geq 0$

$x \geq 0$ & $\sqrt{x} \leq 1$

$\Rightarrow 0 \leq x \leq 1$

Range : $[0, 1]$

10. A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is such that $f\left(\frac{1-x}{1+x}\right) = x$ for all $x \neq -1$. Prove the following :

फलन $f: \mathbb{R} \rightarrow \mathbb{R}$ ऐसा है कि $f\left(\frac{1-x}{1+x}\right) = x$, $x \neq -1$ निम्नलिखित को सिद्ध कीजिये :

(a) $f(f(x)) = x$

(b) $f(1/x) = -f(x)$, $x \neq 0$

(c) $f(-x-2) = -f(x) - 2$

Ans.

Sol. $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f\left(\frac{1-x}{1+x}\right) = x \quad \forall x \in \mathbb{R} - \{-1\}$$

$$y \leftrightarrow \frac{1-x}{1+x} \quad y = \frac{1-x}{1+x}$$

$$\Rightarrow y + xy = 1 - x$$

$$\Rightarrow x(y+1) = 1 - y$$

$$\Rightarrow f(y) = \frac{1-y}{1+y} \quad \Rightarrow x = \frac{1-y}{y+1}$$

$$\Rightarrow f(x) = \frac{1-x}{1+x}$$

$$(a) f(f(x)) = f\left(\frac{1-x}{1+x}\right) = \frac{1 - \left(\frac{1-x}{1+x}\right)}{1 + \frac{1-x}{1+x}} = \frac{2x}{2} = x$$

$$(b) f\left(\frac{1}{x}\right) = \frac{1 - \frac{1}{x}}{1 + \frac{1}{x}} = \frac{x-1}{x+1} = \left(\frac{1-x}{1+x}\right) - 1 = -f(x); x \neq 0$$

$$(c) f(-x-2) = \frac{1 - (-x-2)}{1 + (-x-2)} = \frac{x+3}{-(x+1)}$$

$$= \frac{x-1}{x+1} - 2 = -f(x) - 2$$

11. $f(x) = \begin{cases} 1-x & \text{if } x \leq 0 \\ x^2 & \text{if } x > 0 \end{cases}$ and $g(x) = \begin{cases} -x & \text{if } x < 1 \\ 1-x & \text{if } x \geq 1 \end{cases}$ find $(f \circ g)(x)$ and $(g \circ f)(x)$

$f(x) = \begin{cases} 1-x & \text{if } x \leq 0 \\ x^2 & \text{if } x > 0 \end{cases}$ तथा $g(x) = \begin{cases} -x & \text{if } x < 1 \\ 1-x & \text{if } x \geq 1 \end{cases}$ $(f \circ g)(x)$ तथा $(g \circ f)(x)$ ज्ञात कीजिये।

Ans. $(g \circ f)(x) = \begin{cases} x & \text{if } x \leq 0 \\ -x^2 & \text{if } 0 < x < 1 \\ 1-x^2 & \text{if } x \geq 1 \end{cases}$; $(f \circ g)(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 1+x & \text{if } 0 \leq x < 1 \\ x & \text{if } x \geq 1 \end{cases}$

Sol. $f(x) = \begin{cases} 1-x & x \leq 0 \\ x^2 & x > 0 \end{cases}$ $g(x) = \begin{cases} -x & x < 1 \\ 1-x & x \geq 1 \end{cases}$

$$f \circ g(x) = \begin{cases} 1-g(x) & g(x) \leq 0 \\ (g(x))^2 & g(x) > 0 \end{cases} = \begin{cases} x^2 & x < 0 \\ 1+x & 0 \leq x < 1 \\ x & x \geq 1 \end{cases}$$

$$g \circ f(x) = \begin{cases} -f(x) & f(x) < 1 \\ 1-f(x) & f(x) \geq 1 \end{cases}$$

$$= \begin{cases} x & x \leq 0 \\ -x^2 & 0 < x < 1 \\ 1-x^2 & x \geq 1 \end{cases}$$

12. $f(x)$ and $g(x)$ are linear function such that for all x , $f(g(x))$ and $g(f(x))$ are Identity functions. If $f(0) = 4$ and $g(5) = 17$, compute $f(136)$.

$f(x)$ व $g(x)$ रेखीय फलन है ताकि $\forall x \in \mathbb{R}$, $f(g(x))$ व $g(f(x))$ तत्समक फलन है। यदि $f(0) = 4$, $g(5) = 17$ तो $f(136)$ निकालें।

Ans. 12

Sol. Let $f(x) = ax + b$

$$f(g(x)) = x$$

$$g(x) = cx + d$$

$$\Rightarrow f(cx + d) = x$$

$$\text{Given } f(0) = 4 \text{ \& } g(5) = 17$$

$$\Rightarrow a(cx + d) + b = x$$

$$\Rightarrow b = 4 \text{ \& } 5c + d = 17$$

$$\Rightarrow (ac - 1) = 0 \text{ \& } ad + b = 0$$

$$\text{gof}(x) = x$$

$$\Rightarrow g(ax + b) = x$$

$$\Rightarrow c(ax + b) + d = x$$

$$\Rightarrow ca - 1 = 0 \text{ \& } bc + d = 0$$

$$\Rightarrow ac = 1, b = 4, ad = -4 \text{ \& } d = -4c$$

$$\Rightarrow 5c - 4c = 17$$

$$\Rightarrow c = 17$$

$$\Rightarrow a = \frac{1}{17}$$

$$\therefore f(x) = \frac{x}{17} + 4$$

$$\Rightarrow f(136) = \frac{136}{17} + 4 = 12$$

EVEN AND ODD FUNCTIONS

13. Find whether the following functions are even or odd or none :

ज्ञात करे निम्न फलन सम, विषम या दोनों में से कुछ भी नहीं है :

$$(a) f(x) = \log(x + \sqrt{1+x^2})$$

$$(b) f(x) = \frac{x(a^x + 1)}{a^x - 1}$$

$$(c) f(x) = \sin x + \cos x$$

$$(d) f(x) = x \sin^2 x - x^3$$

$$(e) f(x) = \sin x - \cos x$$

$$(f) f(x) = \frac{(1+2^x)^2}{2^x}$$

$$(g) f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$$

$$(h) f(x) = [(x+1)^2]^{1/3} + [(x-1)^2]^{1/3}$$

Ans. (a) odd (b) even (c) neither odd nor even (d) odd
(e) neither odd nor even (f) even (g) even (h) even

Sol. (a) $f(x) = \log(x + \sqrt{1+x^2})$

$$f(-x) = \log(-x + \sqrt{1+x^2}) = \log\left(\frac{1}{x + \sqrt{1+x^2}}\right)$$

$$= -\log(x + \sqrt{1+x^2}) \rightarrow \text{odd}$$

$$(b) f(x) = \frac{x(a^x + 1)}{a^x - 1}$$

$$f(-x) = \frac{-x(1 + a^x)}{1 - a^x} = f(x)$$

$\Rightarrow$ Even

$$(c) f(x) = \sin x + \cos x$$

$$f(-x) = \cos x - \sin x$$

Neither even nor odd

$$(d) f(x) = x \sin^2 x - x^3 = \underbrace{x(\sin^2 x - x^2)}_{\text{odd}}$$

$$(e) f(x) = \underbrace{\sin x - \cos x}_{\text{Neither even nor odd}}$$

$$(f) f(x) = \frac{(1 + 2^x)^2}{2^x} \rightarrow \text{Even}$$

$$(g) f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1 \rightarrow \text{Even}$$

$$f(-x) = -x \left(\frac{e^x}{1 - e^x} + \frac{1}{2} \right) + 1$$

$$= -x \left(\frac{2e^x + 1 - e^x}{2(1 - e^x)} \right) + 1 = -x \left(\frac{e^x + 1}{2(1 - e^x)} \right) + 1$$

$$= \frac{xe^x + x}{2(e^x - 1)} + 1 = f(x)$$

$$(h) f(x) = \left[(x+1)^2 \right]^{\frac{1}{3}} + \left[(x-1)^2 \right]^{\frac{1}{3}}$$

$$f(-x) = f(x) \rightarrow \text{Even function}$$

INVERSE OF A FUNCTION

14. Compute the inverse of the functions :

निम्न का प्रतिलोम फलन ज्ञात कीजिये :

(a) $f(x) = \ln(x + \sqrt{x^2 + 1})$

(b) $f(x) = 2^{\frac{x}{x-1}}$

(c) $y = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$

Ans. (a) $\frac{e^x - e^{-x}}{2}$ (b) $\frac{\log_2 x}{\log_2 x - 1}$ (c) $\frac{1}{2} \log \frac{1+x}{1-x}$

Sol. (a) $f(x) = \ln(x + \sqrt{x^2 + 1})$

$$y = \ln(x + \sqrt{x^2 + 1})$$

$$x \leftrightarrow y$$

$$x = \ln(y + \sqrt{y^2 + 1})$$

$$e^x = y + \sqrt{y^2 + 1}$$

$$\Rightarrow e^{2x} + y^2 - 2ye^x = y^2 + 1$$

$$\Rightarrow e^{2x} - 1 = 2ye^x$$

$$\Rightarrow y = \frac{e^{2x} - 1}{2e^x} = \frac{e^x - e^{-x}}{2} = f^{-1}(x)$$

(b) $f(x) = 2^{\frac{x}{x-1}}$

$$\Rightarrow y = 2^{\frac{x}{x-1}}$$

$$\Rightarrow x = 2^{\frac{y}{y-1}}$$

$$\Rightarrow \log_2 x = \frac{y}{y-1}$$

$$\Rightarrow y(\log_2 x - 1) = \log_2 x$$

$$\Rightarrow y = f^{-1}(x) = \frac{\log_2 x}{\log_2 x - 1}$$

$$(c) y = \frac{10^x - 10^{-x}}{10^x + 10^{-x}} = \frac{10^{2x} - 1}{10^{2x} + 1}$$

$$\Rightarrow x = \frac{10^{2y} - 1}{10^{2y} + 1}$$

$$\Rightarrow 10^{2y}(x - 1) + x + 1 = 0$$

$$\Rightarrow 10^{2y} = \frac{x + 1}{1 - x}$$

$$\Rightarrow 2y = \log_{10} \left(\frac{x + 1}{1 - x} \right)$$

$$\Rightarrow f^{-1}(x) = \frac{1}{2} \log_{10} \left(\frac{1 + x}{1 - x} \right)$$

15. Function f and g are defined by $f(x) = \sin x$, $x \in \mathbb{R}$; $g(x) = \tan x$, $x \in \mathbb{R} - \left(K + \frac{1}{2} \right) \pi$ where $K \in \mathbb{I}$.

Find (i) periods of $fo g$ and gof . (ii) range of the function $fo g$ and gof .

फलन f तथा g , $f(x) = \sin x$, $x \in \mathbb{R}$; $g(x) = \tan x$, $x \in \mathbb{R} - \left(K + \frac{1}{2} \right) \pi$ के रूप में परिभाषित किया गया है, जहाँ $K \in \mathbb{I}$.

ज्ञात कीजिए (i) $fo g$ व gof का आवर्तकाल ज्ञात कीजिये (ii) $fo g$ व gof परिसर ज्ञात कीजिये।

Ans. (i) period of $fo g$ is π , period of gof is 2π (ii) range of $fo g$ is $[-1, 1]$, range of gof is $[-\tan 1, \tan 1]$

Sol. $fo g(x) = f(\tan x) = \sin(\tan x)$

$$\Rightarrow \underbrace{fo g(\pi + x)}_{P=\pi} = fo g(x)$$

Range of $fo g$ is $[-1, 1]$

$$\underbrace{gof(x)}_{\text{period}=2\pi} = g(\sin x) = \tan(\sin x)$$

Range = $[-\tan 1, \tan 1]$

PERIODIC FUNCTION

16. Suppose that f is an even, periodic function with period 2, and that $f(x) = x$ for all x in the interval $[0, 1]$. Find the value of $f(3.14)$.

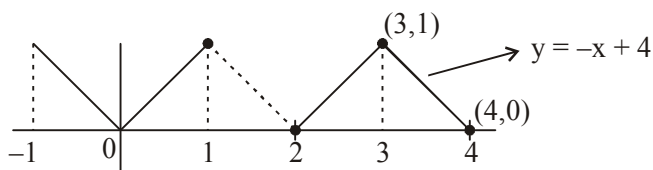
फलन f सम, आवर्ती है जिसका आवर्तकाल 2 है, $f(x) = x$, सभी $x \in [0, 1]$ के लिये। $f(3.14)$ का मान ज्ञात कीजिये।

Ans. 0.86

Sol. $f(-x) = f(x)$

$$f(x+2) = f(x) \quad \forall x \in \mathbb{R}$$

$$f(3.14) = -3.14 + 4 = 0.86$$



17. The fundamental period of $e^{\cos^4 \pi x + x - [x] + \cos \pi x}$ is _____. (where $[]$ denotes the greatest integer function)

$e^{\cos^4 \pi x + x - [x] + \cos \pi x}$ का मूलभूत आवर्तकाल _____ है। (जहाँ $[]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है)

Ans. 2

Sol.
$$e^{\underbrace{\cos^4 \pi x}_{\text{period}=1} + \underbrace{x - [x]}_{\text{period}=1} + \underbrace{\cos \pi x}_{\text{period}=2}}$$

$$\Rightarrow P = \text{L.C.M.}(1, 1, 2) = 2$$

OTHERS

18. If $f(x) = \max(x, 1/x)$ for $x > 0$ where $\max(a, b)$ denotes the greater of the two real numbers a and b . Define the function $g(x) = f(x) \cdot f(1/x)$ and plot its graph.

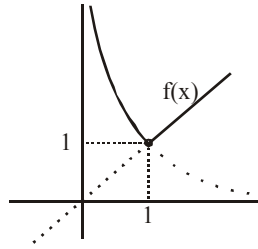
यदि $f(x) = \max(x, 1/x)$, $x > 0$; जहाँ $\max(a, b)$ a व b बड़ी संख्या को निरूपित करता है तो फलन $g(x) = f(x) \cdot f(1/x)$ को परिभाषित करें तथा इसका ग्राफ खींचिए।

Ans.
$$g(x) = \begin{cases} \frac{1}{x^2} & \text{if } 0 < x \leq 1 \\ x^2 & \text{if } x > 1 \end{cases}$$

Sol. $x = \frac{1}{x} \Rightarrow x^2 = 1 \Rightarrow x = 1$

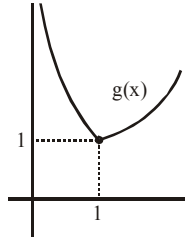
$$f(x) = \frac{1}{x}; \quad 0 < x \leq 1$$

$$= x; \quad x > 1$$



$$g(x) = f(x) f\left(\frac{1}{x}\right)$$

$$\Rightarrow g(x) = \begin{cases} \frac{1}{x} \cdot \frac{1}{x} & 0 < x \leq 1 \\ x \cdot x & x > 1 \end{cases}$$



19. Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$, where $a, b, c, d \in \mathbb{R}$. Suppose $P(0) = 6$, $P(1) = 7$, $P(2) = 8$ and $P(3) = 9$, then find the value of $P(4)$.

माना $P(x) = x^4 + ax^3 + bx^2 + cx + d$, जहाँ $a, b, c, d \in \mathbb{R}$. यदि $P(0) = 6$, $P(1) = 7$, $P(2) = 8$ तथा $P(3) = 9$, तो $P(4)$ का मान ज्ञात कीजिए।

Ans. 34

Sol. Let $g(x) = P(x) - (x + 6)$

$$\Rightarrow g(0) = g(1) = g(2) = g(3) = 0$$

$$\Rightarrow g(x) = x(x-1)(x-2)(x-3)$$

$$\Rightarrow P(x) = x(x-1)(x-2)(x-3) + x + 6$$

$$P(4) = 4 \times 3 \times 2 \times 1 + 10$$

$$= 24 + 10$$

$$= 34$$

20. Let $f(x) = x^{135} + x^{125} - x^{115} + x^5 + 1$. If $f(x)$ is divided by $x^3 - x$ then the remainder is some function of x say $g(x)$. Find the value of $g(10)$.

$f(x) = x^{135} + x^{125} - x^{115} + x^5 + 1$ से विभाजित करने पर शेषफल $g(x)$ है। $g(10)$ का मान ज्ञात कीजिए।

Ans. 21

Sol. $f(x) = x(x-1)(x+1)P(x) + ax^2 + bx + c$

$$\left. \begin{array}{l} f(0) \Rightarrow c = 1 \\ f(1) = a + b + c \Rightarrow a + b + c = 3 \\ f(-1) = a - b + c \Rightarrow a - b + c = -1 \end{array} \right\} \Rightarrow \begin{array}{l} a + b = 2 \\ a - b = -2 \end{array}$$

$$\Rightarrow a = 0, b = 2, c = 1$$

$$\Rightarrow g(x) = 2x + 1$$

$$\Rightarrow g(10) = 21$$

21. Let $f(x) = \frac{9^x}{9^x + 3}$ then find the value of the sum $f\left(\frac{1}{2006}\right) + f\left(\frac{2}{2006}\right) + f\left(\frac{3}{2006}\right) + \dots + f\left(\frac{2005}{2006}\right)$.

$f(x) = \frac{9^x}{9^x + 3}$ तो योग $f\left(\frac{1}{2006}\right) + f\left(\frac{2}{2006}\right) + f\left(\frac{3}{2006}\right) + \dots + f\left(\frac{2005}{2006}\right)$ का मान ज्ञात कीजिए।

Ans. 1002.5

Sol. $f(x) = \frac{9^x}{9^x + 3}$

$$\Rightarrow f(x) + f(1-x) = 1$$

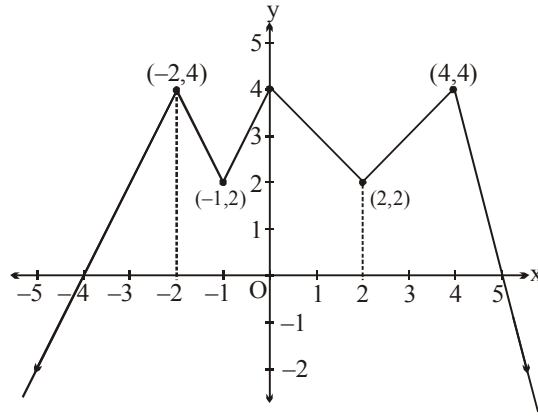
$$\Rightarrow f\left(\frac{1}{2006}\right) + f\left(\frac{2005}{2006}\right) + \dots + f\left(\frac{1002}{2006}\right) + f\left(\frac{1004}{2006}\right) + f\left(\frac{1003}{2006}\right)$$

$$= 1002 + f\left(\frac{1}{2}\right) = 1002 + \frac{\sqrt{9}}{\sqrt{9} + 3} = 1002 + \frac{1}{2}$$

$$= 1002.5$$

22. Consider the graph of a real-valued continuous function $f(x)$ defined on $\mathbb{R}$ (the set of all real numbers) as shown below.

माना $\mathbb{R}$ (सभी वास्तविक संख्याओं का समुच्चय) पर परिभाषित वास्तविक मानों वाले फलन $f(x)$ का आरेख निम्न प्रकार है।



Find the number of real solutions of the equation $f(f(x)) = 4$.

समीकरण $f(f(x)) = 4$ के वास्तविक हलों की संख्या ज्ञात कीजिए।

Ans. 7

Sol. $f(f(x)) = 4$ if $\underbrace{f(x) = 0}_{2 \text{ solution}}$ or $\underbrace{-2}_{2 \text{ solution}}$ or $\underbrace{4}_{3 \text{ solution}}$

$\Rightarrow$ No. of solutions = 7

Paragraph for Question Nos. 23 to 25

Let $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b \quad \forall x \in \mathbb{R}$

माना कि $f(x) = \frac{x^3}{3} + \frac{x^2}{2} + ax + b \quad \forall x \in \mathbb{R}$

23. Least value of 'a' for which $f(x)$ is injective function, is :

$f(x)$ के एकैकी फलन होने के लिए a का न्यूनतम मान है :

- (A) $\frac{1}{4}$ (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{8}$

Ans. A

Sol. $f'(x) = x^2 + x + a \Rightarrow f'(x) \geq 0 \Rightarrow \left(x + \frac{1}{2}\right)^2 + \left(a - \frac{1}{4}\right) \geq 0$

Hence $a \geq \frac{1}{4}$

$f'(x) = x^2 + x + a \Rightarrow f'(x) \geq 0 \Rightarrow \left(x + \frac{1}{2}\right)^2 + \left(a - \frac{1}{4}\right) \geq 0$

अतः $a \geq \frac{1}{4}$

24. If $a = -1$ then $f(x)$ is :
 (A) bijective (B) many-one and onto
 (C) one-one and into (D) many-one and into

यदि $a = -1$ हो, तो $f(x)$ है :

- (A) एकैकी आच्छादक (B) बहुएकैकी और आच्छादक
 (C) एकैकी और अन्तर्क्षेपी (D) बहुएकैकी और अन्तर्क्षेपी

Ans. B

Sol. If $a = -1 \Rightarrow f(x) = x^2 + x - 1$
 $D = 1 + 4 > 0 \Rightarrow$ many one function and cubic polynomial have range $(-\infty, \infty)$
 hence onto function
 यदि $a = -1 \Rightarrow f(x) = x^2 + x - 1$
 $D = 1 + 4 > 0 \Rightarrow$ यह बहुएकैकी फलन है एवं त्रिघात बहुपद का परिसर $(-\infty, \infty)$ होता है।
 अतः यह आच्छादक फलन है।

25. $f(x)$ is invertible iff:

$f(x)$ प्रतिलोमीय है, यदि और केवल यदि :

- (A) $a \in \left[\frac{1}{4}, \infty\right), b \in \mathbb{R}$ (B) $a \in \left[\frac{1}{8}, \infty\right), b \in \mathbb{R}$ (C) $a \in \left(-\infty, \frac{1}{4}\right], b \in \mathbb{R}$ (D) $a \in \left(-\infty, \frac{1}{4}\right), b \in \mathbb{R}$

Ans. A

Sol. For $f(x)$ to be invertible $f(x)$ should be one-one that is $f'(x) \geq 0, \Rightarrow a \in \left[\frac{1}{4}, \infty\right)$

$f(x)$ प्रतिलोमीय हो इसके लिए आवश्यक है कि $f(x)$ एकैकी फलन होना चाहिए

$$\text{अतः } f'(x) \geq 0 \Rightarrow a \in \left[\frac{1}{4}, \infty\right)$$

Paragraph for Question Nos. 26 to 28

Consider the function $y = f(x) = x^2 - 6x + 5$

माना फलन $y = f(x) = x^2 - 6x + 5$

26. The domain of definition of the function $g(x) = \ln(f(x))$:

फलन $g(x) = \ln(f(x))$ का प्रांत होगा :

- (A) $(4, \infty)$ (B) $(-\infty, 1)$ (C) $\mathbb{R} - \{x \mid 1 \leq x \leq 5\}$ (D) $\mathbb{R}$

Ans. C

Sol. $y = x^2 - 6x + 5 = f(x)$

$g(x) = \ln(f(x))$

For D_g $f(x) > 0$

$\Rightarrow x^2 - 6x + 5 > 0$

$\Rightarrow (x-5)(x-1) > 0$

$\Rightarrow x \in (-\infty, 1) \cup (5, \infty)$

$\Rightarrow D_g = \mathbb{R} - \{x \mid 1 \leq x \leq 5\}$

27. Range of the function $y = \ln(f(x))$ is :

फलन $y = \ln(f(x))$ का परिसर होगा :

- (A) $\mathbb{R}$ (B) $[1, \infty)$ (C) $[0, \infty)$ (D) $[-4, \infty)$

Ans. A

Sol. $\because x^2 - 6x + 5 \in [-4, \infty)$

$\Rightarrow \ln(f(x)) \in (-\infty, \infty)$

$\Rightarrow \text{Range of } y = \mathbb{R}$

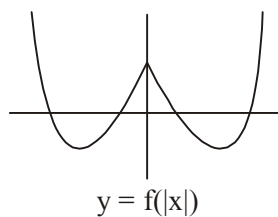
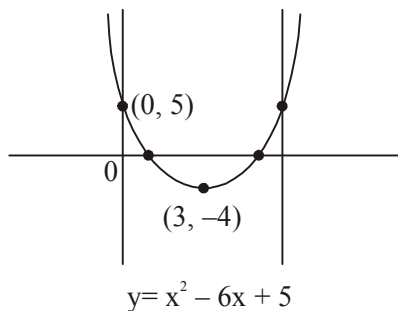
28. Number of points of intersection of a line $y = c$ (where c is a fixed constant and $0 < c < 4$) with graphs $y = |f(|x|)|$ is :

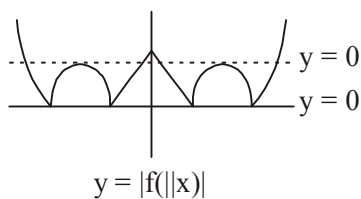
रेखा $y = c$ (जहाँ c एक स्थिर नियतांक है तथा $0 < c < 4$) तथा $y = |f(|x|)|$ के आरेख के प्रतिच्छेद बिन्दु की संख्या होगी :

- (A) 2 (B) 8 (C) 6 (D) 4

Ans. B

Sol. No. of points of intersection of a line $y = c$; $0 < c < 4$ is 8.





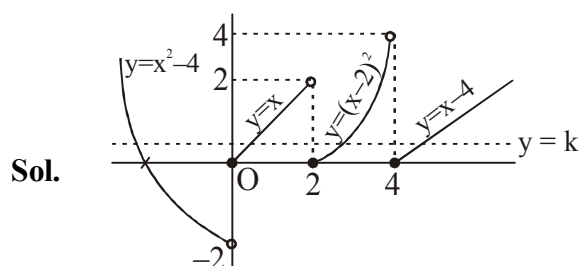
ONE OR MORE THAN ONE CORRECT TYPE

29. Consider $f(x) = \begin{cases} x^2 - 2 & x < 0 \\ x & 0 \leq x < 2 \\ (x-2)^2 & 2 \leq x < 4 \\ x-4 & x \geq 4 \end{cases}$. If the equation $f(x) = k$ has four roots, then k can be equal to :

माना $f(x) = \begin{cases} x^2 - 2 & x < 0 \\ x & 0 \leq x < 2 \\ (x-2)^2 & 2 \leq x < 4 \\ x-4 & x \geq 4 \end{cases}$ है। यदि समीकरण $f(x) = k$ के चार मूल हैं, तो k का मान हो सकता है :

- (A) -2 (B) 0 (C) 1 (D) 2

Ans. BC



So $0 \leq k < 2$

Therefore $k \in \{0, 1\}$

From graph obviously option (B) & (C) are correct.

30. Let $f(x) = x^2 + 4x - 1$ and $g(x) = |x|$. If $h(x) = f(g(x)) + 10$, then the range of $h(x)$, is subset of :

माना $f(x) = x^2 + 4x - 1$ तथा $g(x) = |x|$. यदि $h(x) = f(g(x)) + 10$, तब $h(x)$ का परिसर निम्न में से किसका उपसमुच्चय है :

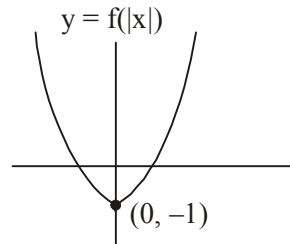
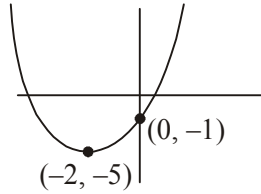
- (A) $\{y : y \geq 10\}$ (B) $\{y : y \geq 0\}$ (C) $\{y : y \geq 5\}$ (D) $\{y : y \geq 9\}$

Ans. BCD

Sol. $h(x) = f(g(x)) + 10 = |x|^2 + 4|x| - 1 + 10 = |x|^2 + 4|x| + 9 = (|x| + 2)^2 + 5$

Hence range of $h(x)$ is $[9, \infty)$

Minimum value occurs when $x = 0$



31. Which of following pairs of functions are identical :

- (A) $f(x) = e^{\ln \sec^{-1} x}$ and $g(x) = \sec^{-1} x$ (B) $f(x) = \tan(\tan^{-1} x)$ & $g(x) = \cot(\cot^{-1} x)$
 (C) $f(x) = \text{sgn}(x)$ and $g(x) = \text{sgn}(\text{sgn}(x))$ (D) $f(x) = \cot^2 x \cdot \cos^2 x$ & $g(x) = \cot^2 x - \cos^2 x$

निम्नलिखित फलनों के युग्मों में से कौनसे तत्समक है :

- (A) $f(x) = e^{\ln \sec^{-1} x}$ and $g(x) = \sec^{-1} x$ (B) $f(x) = \tan(\tan^{-1} x)$ & $g(x) = \cot(\cot^{-1} x)$
 (C) $f(x) = \text{sgn}(x)$ and $g(x) = \text{sgn}(\text{sgn}(x))$ (D) $f(x) = \cot^2 x \cdot \cos^2 x$ & $g(x) = \cot^2 x - \cos^2 x$

Ans. BCD

Sol. (A) $f(x) = e^{\ln(\sec^{-1} x)} = \sec^{-1} x$, $x \in (-\infty, -1] \cup (1, \infty)$
 $g(x) = \sec^{-1} x$, $x \in (-\infty, -1] \cup [1, \infty)$
 non-identical functions तुल्य फलन नहीं है।

(B) $f(x) = \tan(\tan^{-1} x) = x$, $x \in \mathbb{R}$
 $g(x) = \cot(\cot^{-1} x) = x$, $x \in \mathbb{R}$
 identical functions तुल्य फलन है।

(C) $f(x) = \text{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases} \Rightarrow g(x) = \text{sgn}(\text{sgn } x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$

Identical functions तुल्य फलन है।

(D) $f(x) = \cot^2 x \cdot \cos^2 x$, $x \in \mathbb{R} - \{n\pi\}, n \in \mathbb{I}$
 $g(x) = \cot^2 x - \cos^2 x$
 $= \cot^2 x (1 - \sin^2 x) = \cot^2 x \cdot \cos^2 x$ $x \in \mathbb{R} - \{n\pi\}, n \in \mathbb{I}$
 Identical functions तुल्य फलन है।

32. $D \equiv [-1, 1]$ is the domain of the following functions, state which of them are injective :

निम्नलिखित फलनों का प्रान्त $D \equiv [-1, 1]$ है, इनमें से कौनसे फलन एकैकी है :

(A) $f(x) = x^2$

(B) $g(x) = x^3$

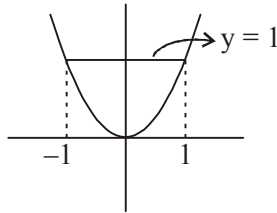
(C) $h(x) = \sin 2x$

(D) $k(x) = \sin(\pi x/2)$

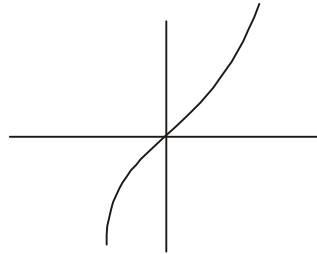
Ans. BD

Sol. (A) $f(x) = x^2$

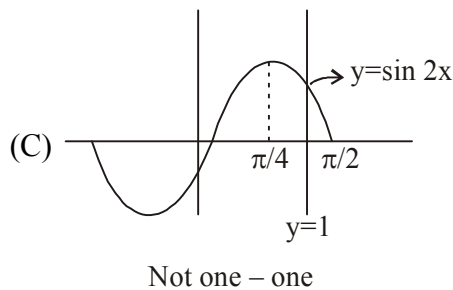
(B) $g(x) = x^3$



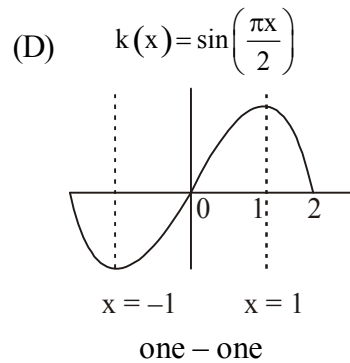
Not one – one



one – one



Not one – one



one – one

33. The functions which are aperiodic are : (where $[x]$ denotes greatest integer function)वे फलन जो अनावर्ती है : (जहाँ $[x]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है।)

(A) $y = [x + 1]$

(B) $y = \sin x^2$

(C) $y = \sin^2 x$

(D) $y = \sin^{-1} x$

Ans. ABD

Sol. (A) $y = [x + 1] \rightarrow$ aperiodic

(B) $y = \sin x^2 \rightarrow$ aperiodic

(C) $y = \sin^2 x \rightarrow$ periodic with period π .

(D) $y = \sin^{-1} x \rightarrow$ aperiodic

34. If $f: \mathbb{R} \rightarrow [-1, 1]$, where $f(x) = \sin \frac{\pi}{2} [x]$, (where $[]$ denotes the greatest integer function) then :

(A) $f(x)$ is onto

(B) $f(x)$ is into

(C) $f(x)$ is periodic

(D) $f(x)$ is many one

यदि $f: \mathbb{R} \rightarrow [-1, 1]$ जहाँ $f(x) = \sin \frac{\pi}{2} [x]$, (जहाँ $[]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है) तो :

- (A) $f(x)$ आच्छादक है (B) $f(x)$ अन्तर्दोषी है (C) $f(x)$ आवर्ती है (D) $f(x)$ बहुएकैकी है

Ans. BCD

Sol. $f: \mathbb{R} \rightarrow [-1, 1] \Rightarrow f(x) = \sin \left(\frac{\pi}{2} [x] \right) = \begin{cases} -1 & , -1 \leq x < 0 \\ 0 & , 0 \leq x < 1 \\ 1 & , 1 \leq x \leq 2 \end{cases}$

Many - one function बहुएकैकी फलन है ।

into function अन्तर्दोषी फलन

Also साथ ही $f(x+4) = \sin \left(\frac{\pi}{2} [x+4] \right) = \sin \left(2\pi + \frac{\pi}{2} [x] \right) = \sin \left(\frac{\pi}{2} [x] \right)$
 $= f(x)$ and hence periodic अतः यह आवृत्ति फलन है ।

35. Identify the statement(s) which is/are incorrect?

- (A) the function $f(x) = \cos (\cos^{-1} x)$ is neither odd nor even
 (B) the fundamental period of $f(x) = \cos (\sin x) + \cos (\cos x)$ is π
 (C) the range of the function $f(x) = \cos (3 \sin x)$ is $[-1, 1]$
 (D) none of these

गलत कथनों की पहचान कीजिए?

- (A) फलन $f(x) = \cos (\cos^{-1} x)$ न तो विषम है न ही सम
 (B) $f(x) = \cos (\sin x) + \cos (\cos x)$ का मूलभूत आवर्तकाल π है
 (C) फलन $f(x) = \cos (3 \sin x)$ का परिसर $[-1, 1]$ है
 (D) इनमें से कोई नहीं

Ans. ABC

Sol. (A) $f(x) = \cos (\cos^{-1} x) = x \rightarrow \text{odd}$

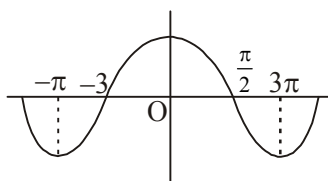
(B) $f(x) = \underbrace{\cos(\sin x)}_{\text{period}=2\pi} + \underbrace{\cos(\cos x)}_{\text{period}=2\pi}$

$\Rightarrow \text{LCM}(2\pi, 2\pi) = 2\pi$

$\Rightarrow \text{Period is } 2\pi.$

(C) $f(x) = \cos \left(\underbrace{3 \sin x}_{\in [-3, 3]} \right)$

$\Rightarrow f(x) \in [\cos 3, 1]$



36. If $F(x) = \frac{\sin \pi [x]}{\{x\}}$, then $F(x)$ is :

- (A) periodic with fundamental period 1
 (B) even
 (C) range is singleton

(D) identical to $\operatorname{sgn} \left(\operatorname{sgn} \frac{\{x\}}{\sqrt{\{x\}}} \right) - 1$, where $\{x\}$ denotes fractional part function and $[.]$ denotes greatest integer function and $\operatorname{sgn}(x)$ is a signum function

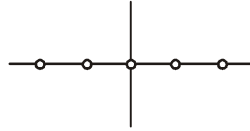
यदि $F(x) = \frac{\sin \pi [x]}{\{x\}}$, हो तो $F(x)$:

- (A) मूलभूत आवर्तकाल 1 का आवर्ती फलन है
 (B) सम फलन है
 (C) एकल परिसर है

(D) $\operatorname{sgn} \left(\operatorname{sgn} \frac{\{x\}}{\sqrt{\{x\}}} \right) - 1$, के समरूप है, जहाँ $\{x\}$ फलन के भिन्नात्मक भाग को प्रदर्शित करता है तथा $[.]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है एवं $\operatorname{sgn}(x)$ एक सिग्नम फलन है।

Ans. ABCD

Sol. $f(x) = \frac{\sin(\pi[x])}{\{x\}} = 0, x \notin I$



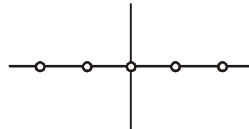
- (A) By graph fundamental period is one
 (B) $f(-x) = 0 = f(x) \therefore$ even function
 (C) Range $y \in \{0\}$

(D) $y = \operatorname{sgn} \left(\operatorname{sgn} \frac{\{x\}}{\sqrt{\{x\}}} \right) - 1, x \notin I$

$$y = \operatorname{sgn}(1) - 1 \Rightarrow y = 1 - 1$$

$y = 0, x \notin I$ Identical to $f(x)$

Hindi $f(x) = \frac{\sin(\pi[x])}{\{x\}} = 0, x \notin I$



- (A) ग्राफ से यह स्पष्ट है कि मूलभूत आवर्तकाल 1 है।
 (B) $f(-x) = 0 = f(x) \therefore$ समफलन
 (C) परिसर $y \in \{0\}$

(D) $y = \operatorname{sgn} \left(\operatorname{sgn} \frac{\{x\}}{\sqrt{\{x\}}} \right) - 1, x \notin I$

$$y = \operatorname{sgn}(1) - 1 \Rightarrow y = 1 - 1$$

$$y = 0, \quad x \notin I \quad f(x) \text{ के समान}$$

37. Function $f(x) = \sin x + \tan x + \operatorname{sgn}(x^2 - 6x + 10)$ is :

- (A) periodic with period 2π (B) periodic with period π
 (C) Non-periodic (D) periodic with period 4π

फलन $f(x) = \sin x + \tan x + \operatorname{sgn}(x^2 - 6x + 10)$ ह :

- (A) 2π आवर्त का आवर्ती फलन (B) π आवर्त का आवर्ती फलन
 (C) अनावर्ती फलन (D) 4π आवर्त का आवर्ती फलन

Ans. AD

Sol. $f(x) = \sin x + \tan x + \underbrace{\operatorname{sgn}(x^2 - 6x + 10)}_{>0}$

$$\Rightarrow f(x) = \underbrace{\sin x}_{P=\pi} + \underbrace{\tan x}_{P=\pi} + \underbrace{1}_{\substack{\downarrow \\ \text{No Fundamental} \\ \text{Period}}}$$

$$\Rightarrow \text{LCM}(2\pi, \pi, \pi) = 2\pi$$

$\Rightarrow f(x)$ is periodic with period 2π or 4π or 6π

38. Let $f: \mathbb{R} \rightarrow \mathbb{R} - \{3\}$ is a function satisfying $f(x + \alpha) = \frac{f(x) - 5}{f(x) - 3}$ for all $x \in \mathbb{R}$, then

- (A) f is periodic with period 4α (B) f is into
 (C) f is constant (D) f can not take the value 1

माना कि $f: \mathbb{R} \rightarrow \mathbb{R} - \{3\}$ एक फलन है जो कि $f(x + \alpha) = \frac{f(x) - 5}{f(x) - 3}$ को x के सभी वास्तविक मानों के लिए संतुष्ट करता है, तो :

- (A) f आवर्ती है तथा 4α आवर्तकाल है (B) f अन्तर्क्षेपी है
 (C) f अचर है (D) f का मान 1 नहीं हो सकता

Ans. ABD

Sol. $f(x + \alpha) = \frac{f(x) - 5}{f(x) - 3}; f(x) \neq 3$

$$f(x+2\alpha) = \frac{f(x+\alpha)-5}{f(x+\alpha)-3} = \frac{\frac{f(x)-5}{f(x)-3}-5}{\frac{f(x)-5}{f(x)-3}-3}$$

$$\Rightarrow f(x+2\alpha) = \frac{-4f(x)+10}{-2f(x)+4} = \frac{2f(x)-5}{f(x)-2}; f(x) \neq 2$$

$$f(x+3\alpha) = \frac{2\left(\frac{f(x)-5}{f(x)-3}\right)-5}{\frac{f(x)-5}{f(x)-3}-2} = \frac{-3f(x)+5}{-f(x)+1}$$

$$\Rightarrow f(x+3\alpha) = \frac{3f(x)-5}{f(x)-1}; f(x) \neq 1$$

$$f(x+4\alpha) = \frac{3\left(\frac{f(x)-5}{f(x)-3}\right)-5}{\frac{f(x)-5}{f(x)-3}-1} = \frac{-2f(x)}{-2} = f(x)$$

$\therefore f(x)$ is periodic with period 4α .

f cannot take the value 1, 2 & 3.

$\Rightarrow f$ is into.

f cannot be constant.

If $f(x) = a$ then

$$a^2 - 3a - a + 5 = 0$$

$$\underbrace{a^2 - 4a + 5 = 0}_{\text{real no. } a}$$

MATCH THE COLUMN

39. Let $f(x) = \ln x$ and $g(x) = x^2 - 1$

Column-I contains composite functions and column-II contains their domain. Match the entries of column-I with their corresponding answer in column-II :

माना $f(x) = \ln x$ तथा $g(x) = x^2 - 1$

स्तम्भ-I में फलन तथा स्तम्भ-II इनके प्रान्त है। सही स्तम्भों का मिलान कीजिए :

Column-I

(A) fog

(B) gof

(C) fof

(D) gog

Column-II

(P) $(1, \infty)$

(Q) $(-\infty, \infty)$

(R) $(-\infty, -1) \cup (1, \infty)$

(S) $(0, \infty)$

Ans. (A) R (B) S (C) P (D) Q

Sol. $f(x) = \ln x$

$g(x) = x^2 - 1$

Domain

$\text{fog}(x) = \ln(x^2 - 1)$

$x \in (-\infty, -1) \cup (1, \infty)$

$\text{gof}(x) = (\ln(x))^2 - 1$

$x \in (0, \infty)$

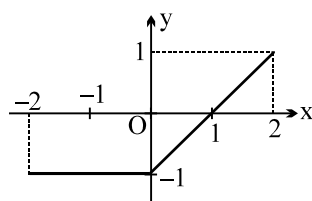
$\text{fof}(x) = \ln(\ln x)$

$x \in (1, \infty)$

$\text{gog}(x) = (x^2 - 1)^2 - 1$

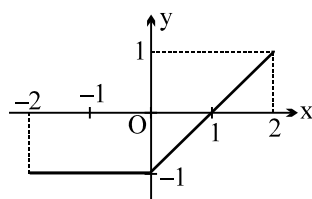
$x \in (-\infty, \infty)$

40. The graph of the function $y = f(x)$ is as follows.



Match the *function* mentioned in Column-I with the respective *graph* given in Column-II.:

$y = f(x)$ का आरेख दिया गया है :



स्तम्भ-I में फलन तथा स्तम्भ-II इनके आरेख है। सही स्तम्भों का मिलान करें।

Column-I

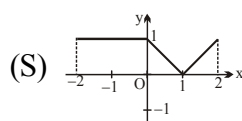
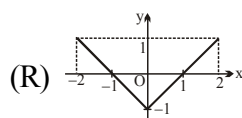
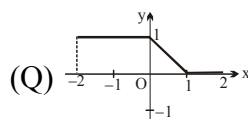
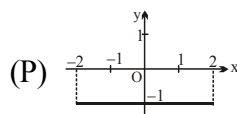
(A) $y = |f(x)|$

(B) $y = f(|x|)$

(C) $y = f(-|x|)$

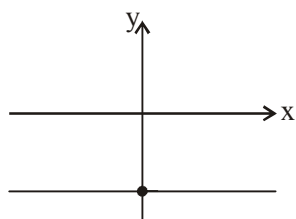
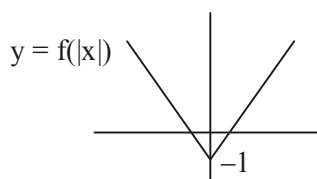
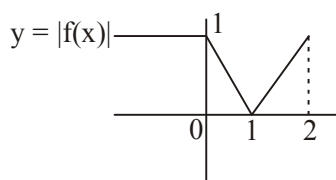
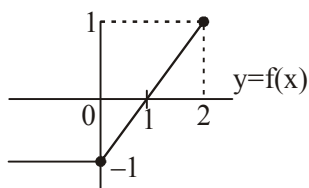
(D) $y = \frac{1}{2} (|f(x)| - f(x))$

Column-II

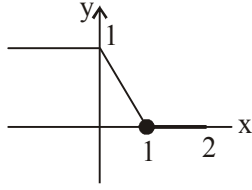


Ans. (A) S (B) R (C) P (D) Q

Sol.



$$y = \frac{1}{2} (|f(x)| - f(x))$$



41. Column I contains functions and column II contains their natural domains. Exactly one entry of column II matches with exactly one entry of column I :

Column I में फलन व column II में प्रान्त है। सही स्तम्भ का मिलाने कीजिए :

Column-I

Column-II

(A) $f(x) = \sin^{-1}\left(\frac{x+1}{x}\right)$

(P) $(1, 3) \cup (3, \infty)$

(B) $g(x) = \sqrt{\ln\left(\frac{x^2 + 3x - 2}{x+1}\right)}$

(Q) $(-\infty, 2)$

(C) $h(x) = \frac{1}{\ln\left(\frac{x-1}{2}\right)}$

(R) $\left(-\infty, -\frac{1}{2}\right]$

(D) $\phi(x) = \ln\left(\sqrt{x^2 + 12} - 2x\right)$

(S) $[-3, -1) \cup [1, \infty)$

Ans. (A) R (B) S (C) P (D) Q

Sol. (A) $f(x) = \sin^{-1}\left(\frac{x+1}{x}\right)$

$$-1 \leq \frac{x+1}{x} \leq 1$$

$$\frac{2x+1}{x} \geq 0$$

$$\frac{1}{x} \leq 0$$

$$x \in \left(-\infty, -\frac{1}{2}\right] \cup (0, \infty)$$

$$x < 0$$

$$x \in \left(-\infty, -\frac{1}{2}\right)$$

$$(B) \ g(x) = \sqrt{\ln\left(\frac{x^2 + 3x - 2}{x + 1}\right)}$$

$$\ln\left(\frac{x^2 + 3x - 2}{x + 1}\right) \geq 0 \quad \& \quad \frac{x^2 + 3x - 2}{x + 1} > 0$$

Taking antilog

$$\frac{x^2 + 3x - 2 - x - 1}{x + 1} \geq 0$$

$$\begin{array}{c} | \quad | \quad | \\ \hline \frac{-3-\sqrt{17}}{2} \quad -1 \quad \frac{-3+\sqrt{17}}{2} \end{array}$$

$$\begin{array}{c} - \quad + \quad - \quad + \\ | \quad | \quad | \quad | \\ \hline -3 \quad -1 \quad 1 \end{array}$$

$$x \in \left(\frac{-3-\sqrt{17}}{2}, -1\right) \cup \left(\frac{-3+\sqrt{17}}{2}, \infty\right)$$

$$\Rightarrow x \in [-3, -1) \cup [1, \infty)$$

$$\Rightarrow \text{Domain} = [-3, -1) \cup [1, \infty)$$

$$(C) \ h(x) = \frac{1}{\ln\left(\frac{x-1}{2}\right)}$$

$$\ln\left(\frac{x-1}{2}\right) \neq 0 \quad \& \quad \frac{x-1}{2} > 0$$

$$\Rightarrow \frac{x-1}{2} \neq 1 \quad \& \quad x > 1$$

$$\Rightarrow x \in (1, 3) \cup (3, \infty)$$

$$(D) \ \phi(x) = \ln\left(\sqrt{x^2 + 12} - 2x\right)$$

$$\sqrt{x^2 + 12} - 2x > 0$$

$$\Rightarrow \sqrt{x^2 + 12} > 2x$$

$$\text{If } x \leq 0 \text{ then it is always true. } \Rightarrow x \in (-\infty, 0] \quad \dots(1)$$

$$\text{If } x > 0 \text{ then } x^2 + 12 > 4x^2$$

$$\Rightarrow 3x^2 < 12$$

$$\Rightarrow x^2 < 4$$

$$\Rightarrow (x - 2)(x + 2) < 0$$

$$\Rightarrow x \in (0, 2) \quad \dots(2)$$

$$\Rightarrow \text{Domain} = (-\infty, 2)$$